Planetary Core Formation via Pebble Accretion:

Disk Truncation Effects

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Disk Truncation Effects

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Abstract

Planetary core formation via pebble accretion has emerged as a leading solution to the timescale problem of classical planetesimal accretion, in which core growth at orbital distances beyond a few astronomical units exceeds the observed lifetime of protoplanetary disks.

In this work, we design, implement, and validate a pebble accretion module within PSyCo, a population synthesis code for planetary systems, based on the analytical prescription of Lambrechts & Johansen (2014). The module integrates the pebble accretion rate alongside the classical Ida & Lin (2008) prescription, allowing a direct quantitative comparison between the two formation pathways under identical disk conditions.

A validation run with four planetary embryos between 2 and 10 AU confirms that pebble-driven cores reach the pebble isolation mass well within the disk lifetime, on timescales ranging from 2.5×10^4 yr at 2 AU to 3.7×10^5 yr at 10 AU, two to three orders of magnitude shorter than classical predictions.

We then examine the effect of disk truncation by varying the outer tapering radius $r_{\text{taper}} \in \{5, 15, 30, 70\}$ AU at fixed total disk mass $M_d = 0.01\ M_\odot$. The results show that more extended disks drive systematically faster core growth at every orbital position, while compact configurations suppress pebble accretion entirely at sufficiently wide orbits, where the local gas surface density falls below the threshold required to sustain pebble-sized particles in the Epstein drag regime.

These results establish that disk geometry governs *when* cores reach the isolation mass, not the mass itself, and that sufficiently compact disks can suppress pebble accretion entirely at wide orbital separations.

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"Science is a choreography . This work is the sum of every hand that steadied me, every voice that reminded me that falling is part of the movement. "

No one arrives at the end of something like this alone. Behind every page there is a history of people who chose, in their own way, to carry part of the weight.

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If I had to distill this journey into a single image, it would be someone who kept looking up even when the ground felt uncertain. Curiosity pulled me forward, and the stubborn refusal to give up kept me there.

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Nomenclature

Acronyms

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PPD	Protoplanetary Disk	1
PSyCo	Planetary Synthesis Code	1

Constants

Notation	Description	Unit
G	Gravitational constant	$6.674 \times 10^{-11} \mathrm{m^3\,kg^{-1}\,s^{-2}}$
γ	Adiabatic index (heat capacity ratio)	1.4
k_B	Boltzmann constant	$1.380\,649 \times 10^{-23} \mathrm{J\,K^{-1}}$
$M_\oplus$	Earth mass	$5.9722 \times 10^{24} \mathrm{kg}$
m_H	Mass of the hydrogen atom	$1.673\,557\,5 \times 10^{-27} \mathrm{kg}$
$M_\odot$	Solar mass	$1.988\,47 \times 10^{30} \mathrm{kg}$
μ	Mean molecular weight of the gas	2.34

Units

Notation	Description	Unit
AU	Astronomical unit (mean Sun–Earth distance)	1.496×10^{13} cm
g cm^{-2}	Surface density unit (CGS)	g cm^{-2}
$L_{\odot}$	Solar luminosity	3.828×10^{33} erg s^{-1}
$M_{\oplus}$	Earth mass	5.972×10^{27} g
M_{J}	Jupiter mass	1.898×10^{30} g
$M_{\odot}$	Solar mass	1.989×10^{33} g
Myr	Million years	3.156×10^{13} s

Symbols & Variables

a	Particle (pebble) radius [cm]
a_{ice}	Ice line radius [AU]
β	Normalization of the Lambrechts & Johansen (2014) gas surface density profile []
β_0	Initial value of β []
c_s	Isothermal sound speed [cm s^{-1}]
η	Sub-Keplerian parameter (dimensionless)
f_d	Disk dust fraction (dimensionless)
$f_{d,0}$	Initial disk dust fraction (dimensionless)
f_g	Disk gas fraction (dimensionless)
$f_{g,0}$	Initial disk gas fraction (dimensionless)
H	Pressure scale height of the disk [cm]
h	Disk aspect ratio H/r (dimensionless)
M_c	Planetary core mass [$M_\oplus$]
$\dot{M}_c$	Core mass accretion rate (pebble accretion) [$M_\oplus$ yr^{-1}]
M_{hydro}	Hydrodynamic critical core mass [$M_\oplus$]
M_{iso}	Planetesimal isolation mass (Ida & Lin) [$M_\oplus$]
$M_{\mathrm{iso,peb}}$	Pebble isolation mass [$M_\oplus$]
M_0	Initial embryo mass [g]
M_g	Accreted gas mass [$M_\oplus$]
$M_{g,\mathrm{iso}}$	Gas isolation mass [$M_\oplus$]
$\dot{M}_{\mathrm{peb}}$	Radial pebble mass flux [g s^{-1}]

m_{peb}	Pebble mass [g]
$m_{\mathrm{peb,min}}$	Minimum pebble mass in the Epstein drag regime [g]
M_t	Drift-to-Hill sphere transition mass [$M_\oplus$]
q	Power-law index of the gas surface density profile
q_s	Power-law index of the solid surface density profile
r	Orbital distance (semi-major axis) [AU]
r_B	Bondi radius [cm]
r_H	Hill radius [cm]
r_{taper}	Outer tapering (truncation) radius of the disk [AU]
Σ_g	Gas surface density []
$\Sigma_{g,0}$	Normalization of the gas surface density at $r_0 =$ 1 AU []
Σ_{peb}	Pebble surface density []
t	Time [yr]
t_f	Friction time [s]
τ_{dep}	Disk gas depletion timescale [Myr]
T	Disk midplane temperature [K]
v_K	Keplerian orbital velocity [$\mathrm{cm\,s^{-1}}$]
v_r	Radial drift velocity [$\mathrm{cm\,s^{-1}}$]
v_{th}	Mean thermal velocity of gas molecules [$\mathrm{cm\,s^{-1}}$]
Z_0	Stellar metallicity, $Z_\star/Z_\odot$ (dimensionless)
Δt	Numerical integration timestep [yr]
η_{ice}	Ice enhancement factor for solid surface density (step function, dimensionless)
ρ_0	Gas midplane volume density [$\mathrm{g\,cm^{-3}}$]
ρ_g	Gas volume density [$\mathrm{g\,cm^{-3}}$]
ρ_s	Internal (bulk) density of solid particles [$\mathrm{g\,cm^{-3}}$]
τ_f	Stokes number (dimensionless)
Ω_K	Keplerian angular frequency [$\mathrm{s^{-1}}$]

1. Introduction

Planetary core formation is one of the fundamental steps in the evolution of planetary systems, and it remains one of the central open problems in modern astrophysics. The classical framework for this process, established by Ida & Lin (2004), models core growth through the gravitational focusing of kilometer-sized planetesimals. However, these models predict formation timescales that frequently exceed the observed lifetime of Protoplanetary Disks (PPDs): at orbital distances beyond ~5 AU, classical accretion timescales reach 10^7–10^8 yr (Lambrechts & Johansen, 2012), two to three orders of magnitude longer than the characteristic disk dispersal timescale $\tau_{\mathrm{dep}} \approx 2.5$ Myr (Mamajek, 2009). Yet gas giants and super-Earths are observed at these separations, most notably the directly imaged planets of HR 8799 at 14–68 AU (Marois et al., 2010) and β Pic b at ~10 AU (Lagrange et al., 2010), implying that a more efficient formation mechanism must operate within the disk lifetime.

Pebble accretion has emerged as a leading solution to this timescale problem (Lambrechts & Johansen, 2012, 2014). By exploiting the aerodynamic drag exerted on millimeter- to centimeter-sized particles in the Epstein regime, planetary embryos can accrete from their full Hill sphere rather than from a geometrically small cross-section, reducing core formation timescales by two to four orders of magnitude at wide orbital separations. The efficiency of this mechanism depends critically on the local gas surface density Σ_g, which is itself set by the radial structure of the disk. In particular, the outer extent of the gas disk, characterized by the tapering radius r_{taper} and shaped by processes such as external photoevaporation (Herrera et al., 2026), directly controls how much gas is available at a given orbital position and therefore how rapidly a core can grow.

The *Planetary Synthesis Code* (Planetary Synthesis Code (PSyCo)) is a planetary population synthesis code developed at Universidad de Antioquia. Its previous implementation modeled core growth exclusively through the classical Ida & Lin planetesimal accretion prescription, lacking a pebble accretion mechanism entirely. This work addresses that gap by designing, implementing, and validating a self-consistent pebble accretion module within PSyCo, based on the analytical prescription of Lambrechts & Johansen (2014). The module operates alongside the existing Ida & Lin prescription, enabling a direct quantitative comparison between the two formation pathways under identical disk conditions within the same simulation environment. This contribution advances PSyCo toward the current state of the art in planet formation modeling and lays the groundwork for future Monte Carlo population synthesis studies.

The results of this work establish three principal findings. First, a validation run with four planetary embryos between 2 and 10 AU confirms that pebble-driven cores reach the pebble isolation mass $M_{\mathrm{iso,peb}}$ on timescales of 2.5×10^4 to 3.7×10^5 yr, two to three orders of magnitude shorter than the classical Ida & Lin prescription under identical conditions. This efficiency opens the possibility of forming not only gas giant cores but also super-Earth mass bodies well within the disk lifetime. Second, a systematic study of disk truncation, varying $r_{\mathrm{taper}} \in \{5, 15, 30, 70\}$ AU at fixed total disk mass, demonstrates that more extended disks supply greater local Σ_g at every orbital position, driving faster core growth. Critically, the pebble isolation mass itself is independent of r_{taper}: all scenarios converge to the same final core mass, differing only in the time required to reach it. Third, in the most compact configuration ($r_{\mathrm{taper}} = 5$ AU), pebble accretion is suppressed entirely at 8 and 10 AU, because the local gas surface density falls below the threshold required to sustain pebble-sized particles in the Epstein drag regime. These results show that disk geometry governs the timescale of pebble-driven core growth, not its asymptotic outcome.

This thesis is organized as follows. Chapter 2 describes the physical structure and temporal evolution of protoplanetary disks, establishing the disk model adopted in PSʏCo. Chapter 3 develops the physics of pebble accretion, covering aerodynamic drag, radial drift, and the transition between the drift and Hill accretion regimes. Chapter 4 derives the analytical prescription for core growth and its adaptation to the PSʏCo disk model. Chapter 5 presents the numerical implementation and the results of the validation run and disk truncation study. Chapter 6 discusses limitations of the current implementation and identifies priorities for future work.

2. Protoplanetary Disk Dynamics

2.1 The Physical Setting and Evolution of Protoplanetary Disks

The physical setting that determines the possibilities of planetary formation begins with the gravitational collapse of a molecular cloud core. A typical solar-mass core possesses a specific angular momentum $l_{\rm core} \sim 4 \times 10^{20}$ cm^2 s^{-1} (Armitage, 2020), a value that exceeds the total angular momentum of the Solar System by three to four orders of magnitude. As a direct consequence of angular momentum conservation, the infalling gas cannot collapse radially onto the central star; instead, it circularizes at a characteristic radius determined by the condition that the Keplerian orbital angular momentum equals that of the progenitor core:

$$\sqrt{GM_* \, r_{\rm circ}} = l_{\rm core} \quad \Longrightarrow \quad r_{\rm circ} = \frac{l_{\rm core}^2}{GM_*} \sim 100 \text{ AU}. \tag{2.1}$$

The formation of circumstellar disks with spatial extents comparable to the Solar System is therefore a geometrically inevitable consequence of the collapse of molecular cloud cores (Armitage, 2020; Williams & Cieza, 2011). According to numerical models of protostellar collapse, these structures consolidate on timescales of order 10^4 yr (Williams & Cieza, 2011), giving rise to the protoplanetary disks observed around young T Tauri stars.

The presence of these disks is detected observationally through the infrared excess emission above the bare stellar photosphere: dust and gas in the disk absorb stellar radiation and re-emit it at longer wavelengths, producing a characteristic signature in the near- and mid-infrared (Williams & Cieza, 2011). Through this diagnostic, statistical surveys of young stellar clusters have traced the temporal evolution of the disk-bearing fraction of stars. The data show that this fraction decays approximately exponentially from values of $\sim$80–90% in clusters of $\sim$ 1–2 Myr to below 10–20% by $\sim$ 10 Myr (Haisch et al., 2001; Mamajek, 2009), with a characteristic timescale

$$f_{\rm disk}(t) \approx f_0 \exp\left(-\frac{t}{\tau_{\rm dep}}\right), \qquad \tau_{\rm dep} \approx 2.5 \text{ Myr}, \tag{2.2}$$

where f_0 is the initial disk fraction at the time of stellar formation (Mamajek, 2009). This disk lifetime is not universal: dispersal proceeds more rapidly around massive stars and in close binary systems (Mamajek, 2009; Williams & Cieza, 2011).

The timescale $\tau_{\rm dep}$ constitutes the fundamental clock against which planetary core formation must compete. Classical planetesimal accretion models (Ida & Lin, 2004) predict formation timescales that frequently exceed several million years, particularly in the outer regions of the disk beyond ~5 AU, where the solid surface density is lower and orbital timescales are longer. This discrepancy between predicted formation times and the available disk lifetime represents the central tension motivating the present work: *what physical mechanism allows massive cores to form before the disk gas disperses?* The answer explored in the following chapters lies in *pebble accretion*, the aerodynamic capture of millimeter- to centimeter-sized particles by planetary embryos, which enables core formation on timescales well within the disk lifetime, as developed quantitatively in Chapter 3.

The disk depletion timescale adopted in the PSʏCo simulations differs from this observational value for physical reasons discussed in Section 2.4.

2.2 Vertical Structure and Characteristic Scales

A protoplanetary disk in steady state must simultaneously satisfy two equilibrium conditions: radial force balance between stellar gravity, centrifugal acceleration, and the gas pressure gradient; and vertical hydrostatic equilibrium between the vertical component of stellar gravity and the thermal pressure of the gas. In this section we derive the vertical structure of the disk, which yields the characteristic scales that appear throughout the analytical prescription developed in Chapter 4.

2.2.1 Hydrostatic Equilibrium in the Vertical Direction

Consider a fluid element of gas located at height z above the disk midplane, at cylindrical distance r from the central star of mass M_*. The vertical component of the stellar gravitational acceleration acting on this element is

$$g_z = -\frac{GM_*}{(r^2 + z^2)} \cdot \frac{z}{\sqrt{r^2 + z^2}}. \tag{2.3}$$

For a geometrically thin disk, the condition $z \ll r$ holds throughout most of the disk volume. Expanding Eq. (2.3) to leading order in z/r yields the *thin-disk approximation*:

$$g_z \approx -\frac{GM_*}{r^3}\, z = -\Omega_K^2\, z, \tag{2.4}$$

where we have defined the **Keplerian angular frequency**

$$\Omega_K \equiv \sqrt{\frac{GM_*}{r^3}}. \tag{2.5}$$

The equation of hydrostatic equilibrium in the vertical direction is then

$$\frac{1}{\rho}\frac{\partial P}{\partial z} = -\Omega_K^2\, z, \tag{2.6}$$

where ρ and P are the local gas density and pressure, respectively. For an ideal gas the equation of state takes the form $P = \rho c_s^2$, where the **isothermal sound speed** is

$$c_s \equiv \sqrt{\frac{k_B T}{\mu m_H}}, \tag{2.7}$$

with k_B the Boltzmann constant, T the local disk temperature, μ the mean molecular weight of the gas ($\mu \approx 2.3$ for a solar-composition mixture dominated by molecular hydrogen), and m_H the proton mass. Substituting $P = \rho c_s^2$ into Eq. (2.6) and assuming that the disk is *vertically isothermal* — that is, $T = T(r)$ depends only on the cylindrical radius — yields a separable ordinary differential equation in z:

$$\frac{1}{\rho}\frac{\partial \rho}{\partial z} = -\frac{\Omega_K^2}{c_s^2}\, z. \tag{2.8}$$

Integrating from the midplane ($z = 0$, $\rho = \rho_0$) to height z:

$$\ln\frac{\rho}{\rho_0} = -\frac{\Omega_K^2}{2c_s^2}\, z^2, \tag{2.9}$$

which gives the well-known Gaussian vertical density profile:

$$\rho(r,z) = \rho_0(r)\exp\left(-\frac{z^2}{2H^2}\right), \tag{2.10}$$

where we have identified the **pressure scale height** of the disk:

$$\boxed{H \equiv \frac{c_s}{\Omega_K}.} \tag{2.11}$$

The scale height H is the most important geometric quantity characterizing the disk: it sets the vertical length scale over which the gas density decreases by a factor $e^{-1/2}$, and it depends on the local temperature through c_s. Warmer disks are geometrically thicker; colder disks are more compressed toward the midplane.

2.2.2 Surface Density and Midplane Density

The **gas surface density** $\Sigma_g(r)$ is defined as the vertically integrated column of gas:

$$\Sigma_g(r) \equiv \int_{-\infty}^{+\infty} \rho(r,z)\,\mathrm{d}z. \tag{2.12}$$

Substituting the Gaussian profile of Eq. (2.10) and using the standard result $\int_{-\infty}^{+\infty} e^{-u^2/2}\,du = \sqrt{2\pi}$:

$$\Sigma_g(r) = \sqrt{2\pi}\,\rho_0(r)\,H. \tag{2.13}$$

Inverting this relation gives the midplane gas density in terms of the surface density:

$$\rho_0(r) = \frac{\Sigma_g(r)}{\sqrt{2\pi}\,H}. \tag{2.14}$$

This expression is particularly useful because Σ_g is the quantity that appears naturally in disk evolution models, while ρ_0 enters directly into the aerodynamic equations governing particle–gas coupling, as we will see in Chapter 3.

2.2.3 Aspect Ratio and Disk Geometry

The **aspect ratio** of the disk is defined as

$$h \equiv \frac{H}{r} = \frac{c_s}{v_K}, \tag{2.15}$$

where $v_K = \sqrt{GM_*/r} = r\Omega_K$ is the Keplerian orbital velocity. The condition $h \ll 1$ is equivalent to requiring that the orbital velocity be highly supersonic, $v_K \gg c_s$, which is well satisfied in all regions of interest for planet formation: typically $h \sim 0.03$–0.1 between 1 and 50 AU.

The radial dependence of h follows directly from the temperature profile of the disk. For a passively irradiated disk, the temperature scales as $T \propto r^{-1/2}$ (derived in Section 2.3), which implies $c_s \propto r^{-1/4}$. Combined with $\Omega_K \propto r^{-3/2}$, the aspect ratio scales as

$$h = \frac{H}{r} \propto \frac{c_s}{\Omega_K\,r} \propto \frac{r^{-1/4}}{r^{-3/2}\cdot r} = r^{1/4}. \tag{2.16}$$

This positive exponent means that the disk *flares* outward: the outer regions subtend a larger solid angle as seen from the star and therefore intercept a greater fraction of the stellar radiation. This mild flaring is self-consistent — a warmer outer disk is geometrically thicker, which in turn leads to greater irradiation — and is a robust prediction of passive disk models (Armitage, 2020). The numerical evaluation of this profile using the MMSN values of Hayashi (1981) is carried out in Section 2.3.

2.3 Temperature Profile and Radial Disk Structure

The spatial distribution of temperature in a protoplanetary disk is set by the dominant heating mechanism. In the inner disk ($r \lesssim 1$–3 AU), viscous dissipation dominates, while in the outer disk ($r \gtrsim 5$ AU) — where giant-planet cores form — stellar irradiation is the primary energy source (Armitage, 2020). We therefore adopt the *passively irradiated disk* model, in which the temperature profile follows from the balance between absorbed stellar radiation and thermal re-emission by the disk surface. This prescription neglects viscous heating and assumes the disk is optically thin to its own thermal emission, conditions that are well satisfied in the outer disk where viscous accretion rates are low (Hayashi, 1981; Ida & Lin, 2008).

Consider a disk surface element at orbital distance r from a star of luminosity L_*. In radiative equilibrium, the absorbed stellar flux equals the blackbody re-emission, $\sigma_{\rm SB}T^4$, which gives

$$T(r) = \left(\frac{\alpha_{\rm irr}\, L_*}{4\pi\sigma_{\rm SB}}\right)^{1/4} r^{-1/2}, \tag{2.17}$$

where $\alpha_{\rm irr}$ is the effective grazing angle of stellar radiation on the flared disk surface and $\sigma_{\rm SB}$ is the Stefan–Boltzmann constant. The $r^{-1/2}$ dependence follows directly from the inverse-square dilution of stellar radiation with distance and is independent of disk structure. Evaluating Eq. (2.17) for standard minimum-mass solar nebula (MMSN) parameters with $M_* = M_\odot$ and $L_* = L_\odot$, Hayashi (1981) obtained

$$\boxed{T(r) = 280\left(\frac{r}{1\ \rm AU}\right)^{-1/2}\left(\frac{L_*}{L_\odot}\right)^{1/4}\ \rm K,} \tag{2.18}$$

valid over the range $0.35 \lesssim r/\rm AU \lesssim 36$ (Hayashi, 1981; Lambrechts & Johansen, 2012). This expression is implemented directly in the disk temperature prescription of PSyCo (function `Td` in `discGen_fd_pebble.py`) and governs all thermodynamic quantities used in the subsequent analysis.

With the temperature profile established, the characteristic scales of the disk follow immediately. The isothermal sound speed $c_s = \sqrt{k_B T/\mu m_H}$ scales as $c_s \propto T^{1/2} \propto r^{-1/4}$, and the pressure scale height $H = c_s/\Omega_K$ as $H \propto r^{5/4}$. The **aspect ratio** of the disk is therefore

$$h \equiv \frac{H}{r} = \frac{c_s}{\Omega_K\, r} \propto \frac{r^{-1/4}}{r^{-3/2} \cdot r} = r^{1/4}, \tag{2.19}$$

which evaluates numerically to (Hayashi, 1981; Lambrechts & Johansen, 2012)

$$\boxed{h = \frac{H}{r} \approx 0.033 \left(\frac{r}{\mathrm{AU}}\right)^{1/4}.} \tag{2.20}$$

The positive exponent confirms that the disk flares outward, self-consistently justifying the passively irradiated model. Since $h \ll 1$ throughout the disk ($h \sim 0.03$–0.10 between 1 and 50 AU), the thin-disk approximation of Section 2.2.1 is also satisfied.

The aspect ratio carries a direct implication for the maximum mass a core can reach by pebble accretion. Pebble accretion halts when the core opens a pressure gap in the disk, which occurs when its Hill radius becomes comparable to the local scale height, $r_H \sim H$. Substituting $r_H = r\,(M_{\rm iso}/3M_*)^{1/3}$ and $H = h \cdot r$, one finds

$$M_{\rm iso,peb} \sim 3M_*\, h^3 \propto r^{3/4}, \tag{2.21}$$

where the exponent $3/4$ emerges directly from the Hayashi profile through the chain $T \propto r^{-1/2} \Rightarrow h \propto r^{1/4} \Rightarrow M_{\rm iso} \propto h^3 \propto r^{3/4}$. Calibrated against hydrodynamical simulations, this yields (Lambrechts & Johansen, 2014)

$$\boxed{M_{\rm iso,peb}(r) \approx 20 \left(\frac{r}{5\,\mathrm{AU}}\right)^{3/4} M_\oplus,} \tag{2.22}$$

giving $\sim$10 $M_\oplus$ at 2 AU and $\sim$33 $M_\oplus$ at 10 AU. This mass plays a dual role in the core accretion scenario: it marks the point at which pebble accretion is abruptly halted by the pressure gap opened in the disk, and simultaneously defines the critical core mass above which the gaseous envelope can no longer be sustained in hydrostatic equilibrium (Lambrechts & Johansen, 2014; Pollack et al., 1996). Once $M_c \geq M_{\rm iso,peb}$, the thermal support of the envelope is removed and rapid runaway gas accretion is triggered, marking the transition from a solid core to a gas giant planet. This expression is implemented in PSyCo as the stopping condition for pebble accretion and is discussed further in Chapter 4.

2.4 Temporal Evolution and Disk Dispersal

The protoplanetary disk does not remain static throughout the planet formation process. The gaseous component evolves and eventually disperses on timescales comparable to those required for planetary core growth, making the temporal evolution of the disk a critical ingredient in any planet formation model. In this section we describe the parametrization adopted in PSyCo for the temporal decay of the gas surface density and discuss its physical consequences for the efficiency of pebble accretion.

Gas surface density evolution

The global depletion of the disk gas is modelled through an exponential decay of the gas fraction $f_g(t)$, following the prescription of Ida & Lin (2008):

$$f_g(t) = f_{g,0}\ \exp\left(-\frac{t}{\tau_{\rm dep}}\right), \tag{2.23}$$

where $f_{g,0} = M_*/M_\odot$ is the initial gas fraction normalized to the minimum-mass solar nebula (MMSN) and $\tau_{\rm dep}$ is the disk depletion timescale, taken in the range 1–10 Myr (Ida & Lin, 2008; Mamajek, 2009). The gas surface density at any radius and time is then

$$\Sigma_g(r,t) = \Sigma_{g,0}\, f_g(t) \left(\frac{r}{1\ \text{AU}}\right)^{-q} \exp\left(-\frac{r}{r_{\rm taper}}\right), \tag{2.24}$$

where $\Sigma_{g,0}$ is the normalization fixed by the total disk mass M_d (see Chapter 5), q is the power-law index of the surface density profile, and $r_{\rm taper}$ is the tapering radius that sets the outer exponential edge of the disk. This profile is implemented directly in the function `Eg` of `discGen_fd_pebble.py`.

The exponential temporal decay in Eq. (2.23) is consistent with the observed statistics of disk fractions in young stellar clusters, which show that the fraction of disk-bearing stars declines with a characteristic e-folding time of $\tau_{\rm dep} \approx 2.5$ Myr (Mamajek, 2009), as discussed in Section 2.1.

Consequences for pebble accretion

The temporal decay of Σ_g has a direct and fundamental consequence for the efficiency of pebble accretion. As derived in Chapter 4, the pebble accretion rate scales as

$$\dot{M}_c \propto \Sigma_g(r,t), \tag{2.25}$$

so that the progressive depletion of the gas surface density naturally quenches core growth over time. This creates a finite window for efficient pebble accretion: cores that do not reach the pebble isolation mass $M_{\rm iso,peb}$ before the gas disperses remain stranded at sub-critical masses and cannot trigger rapid gas accretion. Figure 2.1 illustrates this behavior schematically.

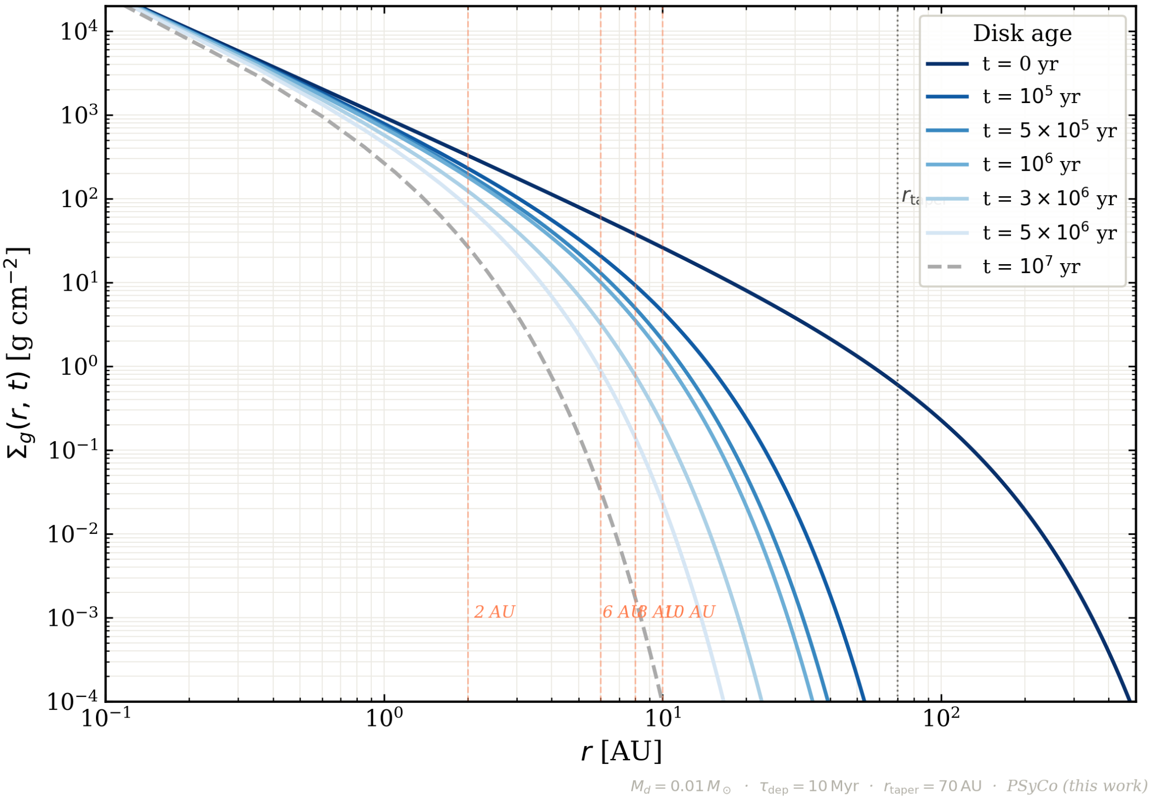

Figure 2.1: Temporal evolution of the gas surface density profile $\Sigma_g(r, t)$ *as a function of orbital distance, computed from the PSyCo disk model (Eq. 2.24). Each curve corresponds to a different disk age, from* $t = 0$ *(darkest blue) to* $t = 10^7$ yr *(dashed grey), illustrating the progressive depletion of the gas reservoir with the exponential decay timescale* $\tau_{\rm dep} = 10$ Myr. *The vertical dashed lines mark the orbital positions of the four embryos used in the validation run (2, 6, 8, and 10 AU), and the dotted vertical line indicates the tapering radius* $r_{\rm taper} = 70$ AU. *Disk parameters:* $M_d = 0.01\,M_\odot$, $\tau_{\rm dep} = 10$ Myr, $r_{\rm taper} = 70$ AU.

The combination of Eqs. (2.23) and (2.24) therefore encodes the fundamental tension introduced in Section 2.1: planetary cores must grow fast enough to reach $M_{\rm iso,peb}$ within $\sim \tau_{\rm dep}$, or else fail to become gas giants. This is precisely the timescale problem that pebble accretion is designed to resolve.

In addition to the smooth exponential decay described above, the outer disk can be further truncated by external photoevaporation in highly irradiated star-forming environments. The implementation of this process and its consequences for planetary populations in such environments is addressed in the companion work by Herrera et al. (2026).

2.5 The Timescale Problem in Planet Formation

The physical framework developed in the preceding sections reveals a fundamental tension at the heart of planet formation theory. Protoplanetary disks are transient structures that disperse on timescales of a few million years. The disk provides both the raw material for planetary growth and the aerodynamic environment that makes pebble accretion possible. Once the gas dissipates, both of these conditions cease to exist.

The severity of this constraint becomes clear when the relevant timescales are compared directly. The characteristic disk lifetime is $\tau_{\rm dep} \approx 2.5$ Myr (Mamajek, 2009), while classical planetesimal accretion models predict core formation timescales that exceed this value by factors of several, particularly in the outer disk beyond $\sim$5 AU where orbital periods are long and solid surface densities are low (Ida & Lin, 2004; Pollack et al., 1996). At 10 AU, for instance, classical accretion timescales can reach 10^7–10^8 yr (Lambrechts & Johansen, 2012), two to three orders of magnitude longer than the available window. Yet gas giants are observed at these separations, most notably the planets of HR 8799 at 14–68 AU (Marois et al., 2010) and β Pic b at $\sim$10 AU (Lagrange et al., 2010), implying that an efficient formation mechanism must operate on much shorter timescales.

The gas surface density profile derived in Section 2.4 makes this constraint quantitative. Since $\dot{M}_c \propto \Sigma_g(r,t)$ and Σ_g decays exponentially with timescale $\tau_{\rm dep}$, the integrated solid mass accreted by a core over the disk lifetime is finite and strongly dependent on the initial conditions. Cores that grow slowly, as in classical planetesimal accretion, never accumulate enough mass to trigger the hydrodynamic collapse of their gaseous envelopes before the disk disperses. The formation of gas giants is therefore not merely a matter of having enough solid material in the disk; it requires that the accretion mechanism be intrinsically rapid.

This is precisely the problem that the pebble accretion framework, introduced by Lambrechts & Johansen (2012) and developed analytically in Lambrechts & Johansen (2014), is designed to resolve. By exploiting the aerodynamic coupling between millimeter- to centimeter-sized particles and the disk gas, pebble accretion achieves growth rates that are two to four orders of magnitude faster than classical planetesimal accretion at the same orbital distance (Lambrechts & Johansen, 2012). The physical basis of this mechanism is developed in Chapter 3, which provides the theoretical foundation for the analytical prescription implemented in PSyCo and described in Chapter 4.

3. Physics of Pebble Accretion

3.1 What is a Pebble?

In the context of planet formation, the term *pebble* does not refer to a geological size classification but to a dynamically defined population of solid particles whose aerodynamic properties place them in a regime of particularly efficient interaction with both the disk gas and growing planetary embryos. Specifically, pebbles are millimeter- to centimeter-sized particles with dimensionless friction times $\tau_f \sim 0.01 - 1$ a range in which aerodynamic drag is strong enough to dissipate kinetic energy during close encounters with a core, yet weak enough to allow significant radial drift relative to the gas (Lambrechts & Johansen, 2012; Armitage, 2020).

Dust grains with $\tau_f \ll 1$ are so tightly coupled to the gas that they follow the gas streamlines and are effectively swept past a growing embryo without being accreted. Kilometer-sized planetesimals, on the other hand, are so decoupled from the gas ($\tau_f \gg 1$) that aerodynamic drag plays no significant role in their capture, and accretion must rely on direct gravitational focusing alone (Lambrechts & Johansen, 2012; Ormel & Klahr, 2010). Pebbles occupy the sweet spot between these two extremes: they are aerodynamically responsive enough to lose energy through gas drag during a close encounter with a core, yet sufficiently decoupled to drift radially across the disk and continuously replenish the feeding zone of a growing embryo (Johansen & Lambrechts, 2017).

The physical size corresponding to this dynamical regime depends on the local gas density and therefore on the orbital distance. In the giant planet formation zone between 5 and 50 AU, friction times of $\tau_f \sim 0.01$–0.1 correspond to particle radii of roughly 1 mm to 1 cm (Lambrechts & Johansen, 2012), consistent with the grain sizes inferred from millimeter and centimeter wavelength continuum observations of protoplanetary disks (Testi et al., 2014). The observational evidence therefore supports the existence of a large reservoir of pebble-sized particles in the outer disk during the epoch of planet formation, providing the raw material required for rapid core growth.

A key property of pebbles is that they do not accumulate locally but drift radially inward due to the aerodynamic headwind they experience from the sub-Keplerian gas. This drift, derived quantitatively in Section 3.3, means that the pebble population in the inner disk is continuously supplied from a large reservoir of growing dust in the outer disk. The resulting radial mass flux of pebbles is the quantity that ultimately controls the rate of core growth in the pebble accretion framework (Lambrechts & Johansen, 2014).

Finally, it is important to distinguish pebble accretion from classical planetesimal accretion not merely as a quantitative improvement but as a qualitatively different physical mechanism. In classical models, core growth proceeds through gravitational focusing of kilometer-sized bodies whose random velocities are excited by mutual interactions, leading to growth timescales that increase steeply with orbital distance and exceed the disk lifetime beyond ~5 AU (Ida & Lin, 2004). Pebble accretion instead exploits the aerodynamic dissipation of energy during particle–core encounters, allowing cores to accrete from the full Hill sphere rather than from a geometrically small cross-section. As demonstrated by Lambrechts & Johansen (2012), this reduces the core formation timescale by two to four orders of magnitude at wide orbital separations, resolving the timescale problem introduced in Section 2.5.

3.2 Gas Drag and Aerodynamic Coupling

The motion of solid particles in a protoplanetary disk is governed by the gravitational field of the central star and by the aerodynamic drag force exerted by the surrounding gas. For particles smaller than the mean free path of the gas molecules, this interaction is described by the *Epstein drag regime* (Epstein, 1924), in which the drag force is proportional to the relative velocity between the particle and the gas:

$$\mathbf{F}_{\rm drag} = -\frac{m}{t_f}\left(\mathbf{v}_{\rm p} - \mathbf{v}_{\rm g}\right), \tag{3.1}$$

where $\mathbf{v}_{\rm p}$ and $\mathbf{v}_{\rm g}$ are the particle and gas velocities respectively, m is the particle mass, and t_f is the **friction time** defined as the characteristic timescale over which the drag force damps the relative velocity between the particle and the gas. In the Epstein regime, the friction time is given by

$$t_f = \frac{\rho_s\, a}{\rho_g\, v_{\rm th}}, \tag{3.2}$$

where ρ_s is the internal density of the particle, a is its radius, ρ_g is the local gas density, and $v_{\rm th} = \sqrt{8k_BT/\pi\mu m_H}$ is the mean thermal velocity of the gas molecules. The Epstein regime is valid when the particle radius satisfies $a \leq (9/4)\lambda$, where λ is the mean free path of the gas (Lambrechts & Johansen, 2012). In the giant planet formation zone between 5 and 50 AU, this condition is well satisfied for millimeter- to centimeter-sized particles, confirming that pebbles in this region are correctly described by Epstein drag.

The degree of aerodynamic coupling between a particle and the gas is characterized by the dimensionless **Stokes number**

$$\tau_f \equiv t_f \, \Omega_K, \tag{3.3}$$

which measures the ratio of the friction time to the orbital timescale Ω_K^{-1}. This single parameter controls the dynamical behavior of the particle in the disk:

- $\tau_f \ll 1$: the particle is strongly coupled to the gas, follows gas streamlines closely, and experiences negligible drift relative to the gas. This corresponds to small dust grains.
- $\tau_f \sim 1$: the particle is optimally decoupled from the gas, experiences maximum radial drift, and loses energy most efficiently during gravitational encounters with a core. This is the pebble regime.
- $\tau_f \gg 1$: the particle is essentially decoupled from the gas, behaves as a test particle in the gravitational field of the star, and is only weakly affected by aerodynamic forces. This corresponds to large planetesimals.

For the pebble accretion mechanism to operate efficiently, particles must have friction times comparable to or shorter than the gravitational deflection time during a close encounter with the core. When this condition is satisfied, gas drag dissipates enough kinetic energy during the encounter to bind the particle to the core, leading to accretion. When it is not satisfied the particle is gravitationally scattered rather than captured (Ormel & Klahr, 2010; Lambrechts & Johansen, 2012).

The quantitative condition that a particle must satisfy to remain in the Epstein regime and participate in pebble accretion is derived in Chapter 4 and implemented in PSyCo through the function `m_pebble_min` in `discGen_fd_pebble.py`.

3.3 Radial Drift of Pebbles

The continuous supply of pebbles to the inner disk depends on the radial drift of solid particles driven by the aerodynamic interaction with the disk gas. In this section we derive the drift velocity of pebbles and the resulting radial mass flux, which is the global quantity that controls core growth in the framework of Lambrechts & Johansen (2014).

3.3.1 The Sub-Keplerian Gas Velocity

The gas in a protoplanetary disk does not orbit at the pure Keplerian velocity. The radially outward pressure gradient in the disk provides a partial support against stellar gravity, reducing the effective gravitational acceleration and allowing the gas to orbit at a slightly sub-Keplerian velocity. The fractional deviation from Keplerian rotation is characterized by the dimensionless parameter

$$\eta \equiv -\frac{1}{2}\left(\frac{H}{r}\right)^2 \frac{\partial \ln P}{\partial \ln r}, \tag{3.4}$$

where $P = \rho_g c_s^2$ is the gas pressure (Nakagawa et al., 1986). For a disk with the Hayashi temperature profile and a power-law surface density $\Sigma_g \propto r^{-1}$, this evaluates to

$$\eta \approx 0.0015 \left(\frac{r}{\mathrm{AU}}\right)^{1/2}, \tag{3.5}$$

so that the azimuthal gas velocity is $v_{\phi,g} = (1-\eta)v_K$, where $v_K = \sqrt{GM_*/r}$ is the Keplerian velocity (Lambrechts & Johansen, 2014). The quantity ηv_K defines the **headwind** that a particle on a Keplerian orbit experiences from the sub-Keplerian gas. For typical disk conditions, $\eta v_K \sim 30$–$50\ \mathrm{m\ s^{-1}}$, a velocity small compared to v_K but large enough to drive significant radial drift on timescales much shorter than the disk lifetime.

3.3.2 Radial Drift Velocity

A solid particle on a Keplerian orbit experiences a continuous headwind from the sub-Keplerian gas. This headwind exerts a drag force that removes angular momentum from the particle, causing it to spiral inward. The equilibrium radial and azimuthal drift velocities of a particle with Stokes number τ_f are (Weidenschilling, 1977; Nakagawa et al., 1986):

$$v_r = -\frac{2\tau_f}{\tau_f^2 + 1}\eta v_K, \tag{3.6}$$

$$v_\phi = -\frac{1}{\tau_f^2 + 1}\eta v_K. \tag{3.7}$$

In the limit $\tau_f \ll 1$, which applies to pebbles in the giant planet formation zone, these simplify to

$$v_r \approx -2\tau_f\, \eta v_K, \tag{3.8}$$

showing that the radial drift velocity is proportional to the Stokes number. Particles with $\tau_f \sim 0.1$ at 10 AU drift inward on a timescale of

$$t_{\mathrm{drift}} = \frac{r}{|v_r|} \approx 550 \left(\frac{\tau_f}{0.1}\right)^{-1} \left(\frac{r}{\mathrm{AU}}\right)\ \mathrm{yr}, \tag{3.9}$$

which is two to three orders of magnitude shorter than the disk lifetime (Lambrechts & Johansen, 2014). This rapid inward drift is what continuously replenishes the feeding zone of a growing core from the outer pebble reservoir.

3.3.3 Radial Pebble Flux

The radial mass flux of pebbles through the disk at orbital distance r is obtained by integrating the drift velocity over the pebble column density:

$$\dot{M}_{\rm peb} = 2\pi r\, \Sigma_{\rm peb}\, |v_r|, \tag{3.10}$$

where $\Sigma_{\rm peb}$ is the pebble surface density. Substituting the drift velocity from Eq. (3.8) and using the pebble surface density derived by Lambrechts & Johansen (2014) from a model of dust coagulation and radial transport:

$$\Sigma_{\rm peb}(r,t) \approx 0.069 \left(\frac{\beta}{500\,{\rm g\,cm^{-2}}}\right) \left(\frac{Z_0}{0.01}\right)^{5/6} \left(\frac{M_*}{M_\odot}\right)^{-1/12} \left(\frac{t}{10^6\,{\rm yr}}\right)^{-1/6} \left(\frac{r}{10\,{\rm AU}}\right)^{-3/4} {\rm g\,cm^{-2}}, \tag{3.11}$$

the resulting pebble flux shows only a weak dependence on time and orbital distance, remaining of order $\dot{M}_{\rm peb} \sim 10^{-4}\,M_\oplus\,{\rm yr}^{-1}$ throughout most of the disk lifetime (Lambrechts & Johansen, 2014). This sustained flux is what makes pebble accretion efficient over million-year timescales: even as individual pebbles drift inward and are lost to the inner disk or sublimation fronts, new pebbles are continuously produced in the outer disk by dust coagulation, maintaining a quasi-steady supply.

It is this global pebble flux that ultimately determines the rate at which a planetary core grows. The connection between $\dot{M}_{\rm peb}$, $\Sigma_{\rm peb}$, and the analytical accretion rate $\dot{M}_c$ is made explicit in Chapter 4, where the full scaling relation for core growth is derived and adapted to the disk model of PSyCo.

3.4 Accretion Regimes: Drift and Hill

The physical mechanism by which a growing planetary core captures pebbles depends critically on the mass of the core relative to a characteristic transition mass. As demonstrated numerically by Lambrechts & Johansen (2012), pebble accretion operates in two distinct regimes each governed by different length scales and accretion rates. Understanding this transition is essential for interpreting the analytical prescription implemented in PSyCo, which operates exclusively in the Hill regime for the core masses of interest in this work.

3.4.1 The Hill Radius and the Bondi Radius

Two characteristic length scales govern the interaction between a growing core and the surrounding pebble population. The **Hill radius**

$$r_H = r\left(\frac{M_c}{3M_*}\right)^{1/3} \tag{3.12}$$

defines the region within which the gravitational attraction of the core dominates over the tidal field of the central star (Lambrechts & Johansen, 2012). Pebbles entering the Hill sphere experience a gravitational deflection on a timescale $\sim \Omega_K^{-1}$, independent of core mass. The Hill radius grows with both core mass and orbital distance as $r_H \propto M_c^{1/3} r$, placing progressively more material within the gravitational reach of the core as it grows.

The **Bondi radius**

$$r_B = \frac{GM_c}{\Delta v^2} \tag{3.13}$$

defines the region within which a particle approaching with relative velocity $\Delta v \approx \eta v_K$ is significantly deflected by the core's gravity (Lambrechts & Johansen, 2012). Unlike the Hill radius, the Bondi radius scales as $r_B \propto M_c$, growing more rapidly with core mass in the low-mass regime. The transition between the two accretion regimes occurs when $r_B \sim r_H$, defining the **transition mass**

$$M_t \approx \frac{1}{3} \frac{(\eta v_K)^3}{G\Omega_K} \approx 3 \times 10^{-3} \left(\frac{\Delta}{0.05}\right)^3 \left(\frac{r}{5\,\mathrm{AU}}\right)^{3/4} M_\oplus, \tag{3.14}$$

where $\Delta = \eta v_K / c_s$ is the dimensionless headwind parameter (Lambrechts & Johansen, 2012). For typical disk conditions, $M_t \sim 10^{-3}$–$10^{-2}\, M_\oplus$, corresponding to bodies of roughly 1000 km in radius.

3.4.2 The Drift Regime

For core masses $M_c < M_t$, the Bondi radius is smaller than the Hill radius. Pebbles approach the core with a mean velocity set by the sub-Keplerian headwind, $\Delta v \approx \eta v_K$, and are accreted within a radius $r_d \leq r_B$. The accretion rate in this regime scales as $\dot{M}_c \propto M_c^2$, faster than exponential, but the absolute rates are low because the effective capture cross-section is small compared to the Hill sphere (Lambrechts & Johansen, 2012). Growth in the drift regime is therefore slow and sensitive to the local headwind: regions of reduced pressure gradient (low η), such as near pressure bumps, can significantly accelerate drift-regime accretion.

In practice, the drift regime is relevant only for the earliest phase of core growth, from the initial seed mass up to M_t. For the planetary embryos considered in this work, with initial masses of $\sim 0.01\, M_\oplus$ (Section 2.1), this transition is reached rapidly and the subsequent evolution is dominated by Hill accretion.

3.4.3 The Hill Regime

For core masses $M_c > M_t$, the Hill radius exceeds the Bondi radius and the Keplerian shear across the Hill sphere dominates over the headwind velocity. In this regime, pebbles entering the Hill sphere approach the core with a characteristic velocity

$$v_H = \Omega_K\, r_H, \tag{3.15}$$

and the gas drag timescale becomes comparable to the crossing time Ω_K^{-1}, allowing all pebbles within the Hill sphere to be captured efficiently (Lambrechts & Johansen, 2012). The accretion rate in the Hill regime is

$$\dot{M}_H = 2\, r_H\, \Sigma_{\rm peb}\, v_H \propto M_c^{2/3}, \tag{3.16}$$

where the $M_c^{2/3}$ scaling reflects the growth of the Hill radius with core mass. This accretion rate is independent of the headwind parameter Δ, making Hill accretion robust to variations in the local pressure gradient (Lambrechts & Johansen, 2012).

The growth timescale in the Hill regime at 5 AU is remarkably short:

$$\Delta t_H \approx 4 \times 10^4 \left(\frac{M_{\rm crit}}{10\, M_\oplus}\right)^{1/3} \left(\frac{r}{5\,\rm AU}\right)\ \rm yr, \tag{3.17}$$

and scales *linearly* with orbital distance, as opposed to the quadratic scaling of drift accretion (Lambrechts & Johansen, 2012). This linear scaling is what makes Hill accretion efficient even at large orbital separations, and is the key property that resolves the timescale problem discussed in Section 2.5.

The full analytical scaling relation for $\dot{M}_c$ in the Hill regime, adapted to the disk model of PSyCo and incorporating the temporal evolution of $\Sigma_g(r,t)$, is derived in Chapter 4. The implementation of both the accretion rate and the transition between regimes in the numerical integrator is described in Chapter 5.

4. Analytical Prescription for Core Growth

This chapter derives the analytical prescription for the growth of planetary cores through pebble accretion. Based on the theoretical framework established by Lambrechts & Johansen (2014), we formulate the mass accretion rate $\dot{M}_c$ as a scaling relation directly connected to the radial flux of pebbles and the Stokes number τ_f, which encodes the aerodynamic coupling of solids within the protoplanetary disk. The transition between 2D and 3D accretion regimes and the physical constraints on core mass are discussed in detail. These analytical expressions are integrated with the disk surface density prescription of PSYCO to resolve the timescale problem inherent in classical models, providing the mathematical foundation implemented in Chapter 5.

4.1 Derivation of the Mass Growth Rate ($\dot{M}_c$)

Planetary core growth through pebble accretion is described by the analytical scaling relations of Lambrechts & Johansen (2014). In gas-rich protoplanetary disks, aerodynamic drag acting on millimeter- to centimeter-sized particles enables efficient capture by planetary embryos, producing growth timescales significantly shorter than those of classical planetesimal accretion (Ida & Lin, 2004). In the Hill accretion regime (Section 3.4.3), the core mass growth rate is approximated by

$$\begin{aligned}\dot{M}_c \approx 4.8 \times 10^{-6} \left(\frac{M_c}{M_\oplus}\right)^{2/3} \left(\frac{Z_0}{0.01}\right)^{25/18} \left(\frac{M_*}{M_\odot}\right)^{-11/36} \\ \times \left(\frac{\beta}{500\,\mathrm{g\,cm^{-2}}}\right) \left(\frac{r}{10\,\mathrm{AU}}\right)^{-5/12} \left(\frac{t}{10^6\,\mathrm{yr}}\right)^{-5/18} \quad M_\oplus\,\mathrm{yr}^{-1},\end{aligned} \tag{4.1}$$

where M_c is the planetary core mass, Z_0 the stellar metallicity normalized to solar, M_* the stellar mass, β the normalization constant of the gas surface density profile, r the orbital distance, and t the disk age. The $M_c^{2/3}$ scaling reflects the growth of the Hill radius with core mass (Eq. 3.12), while the temporal decay $t^{-5/18}$ encodes the progressive depletion of the pebble surface density as the disk evolves.

4.2 Adaptation to the PSyCo Disk Model

The prescription of Lambrechts & Johansen (2014) assumes a gas surface density of the form

$$\Sigma_g(r,t) = \beta \left(\frac{r}{\mathrm{AU}}\right)^{-1}, \qquad \beta = \beta_0\, e^{-t/\tau_{\mathrm{dep}}}, \tag{4.2}$$

where β_0 is the initial normalization. In this work we adopt the disk prescription described in Section 2.4:

$$\Sigma_g(r,t) = \Sigma_{g,0}\, e^{-t/\tau_{\mathrm{dep}}} \left(\frac{r}{1\,\mathrm{AU}}\right)^{-1} \exp\!\left(-\frac{r}{r_{\mathrm{taper}}}\right), \tag{4.3}$$

which is identical in functional form to the Lambrechts & Johansen (2014) profile when evaluated at $r \ll r_{\mathrm{taper}}$, with $\Sigma_{g,0}$ playing the role of β_0. To rewrite Eq. (4.1) in terms of $\Sigma_g(r,t)$, we combine the radial dependences:

$$\left(\frac{\beta}{500\,\mathrm{g\,cm^{-2}}}\right)\left(\frac{r}{10\,\mathrm{AU}}\right)^{-5/12} = \frac{\beta}{500\,\mathrm{g\,cm^{-2}}}\left(\frac{r}{10\,\mathrm{AU}}\right)^{7/12}\left(\frac{r}{10\,\mathrm{AU}}\right)^{-1}. \tag{4.4}$$

Using the identity $(r/10\,\mathrm{AU})^{-1} = 10\,(r/1\,\mathrm{AU})^{-1}$ and substituting $\beta = \Sigma_{g,0}\, e^{-t/\tau_{\mathrm{dep}}}$:

$$= \left(\frac{r}{10\,\mathrm{AU}}\right)^{7/12}\left(\frac{\Sigma_g(r,t)}{50\,\mathrm{g\,cm^{-2}}}\right). \tag{4.5}$$

Substituting into Eq. (4.1), the adapted accretion rate for the PSyCo disk model becomes

$$\begin{aligned}\dot{M}_c \approx 4.8\times 10^{-6}\left(\frac{M_c}{M_\oplus}\right)^{2/3}\left(\frac{Z_0}{0.01}\right)^{25/18}\left(\frac{M_*}{M_\odot}\right)^{-11/36} \\ \times\left(\frac{r}{10\,\mathrm{AU}}\right)^{7/12}\left(\frac{\Sigma_g(r,t)}{50\,\mathrm{g\,cm^{-2}}}\right)\left(\frac{t}{10^6\,\mathrm{yr}}\right)^{-5/18} \quad M_\oplus\,\mathrm{yr}^{-1}.\end{aligned} \tag{4.6}$$

This is the expression implemented in PSyCo through the function `Mcore_pebble_dot` in `discGen_fd_pebble.py`. The change in the radial exponent from $-5/12$ to $+7/12$ reflects the explicit incorporation of the local gas surface density: since $\Sigma_g(r,t) \propto r^{-1}$, the remaining positive power of r implies that pebble accretion becomes comparatively more efficient at larger orbital separations for fixed core mass and metallicity. Furthermore, the exponential temporal decay of $\Sigma_g(r,t)$ naturally quenches the accretion rate as the disk disperses, defining a finite formation window consistent with the observed disk lifetimes discussed in Section 2.4.

4.3 Radial Flux of Pebbles

The efficiency of pebble accretion depends not only on the local properties of the planetary core but also on the global supply of solids drifting inward through the disk. As derived in Section 3.3, the radial mass flux of pebbles at orbital distance r is

$$\dot{M}_{\rm peb} = 2\pi r\, \Sigma_{\rm peb}\, |v_r|, \tag{4.7}$$

where $\Sigma_{\rm peb}$ is the pebble surface density and $v_r \approx -2\tau_f \eta v_K$ is the radial drift velocity (Eq. 3.8). The pebble surface density evolves as (Lambrechts & Johansen, 2014)

$$\Sigma_{\rm peb}(r,t) \approx 0.069 \left(\frac{\beta}{500\,{\rm g\,cm^{-2}}}\right) \left(\frac{Z_0}{0.01}\right)^{5/6} \left(\frac{M_*}{M_\odot}\right)^{-1/12} \left(\frac{t}{10^6\,{\rm yr}}\right)^{-1/6} \left(\frac{r}{10\,{\rm AU}}\right)^{-3/4} {\rm g\,cm^{-2}}, \tag{4.8}$$

yielding a total flux of order $\dot{M}_{\rm peb} \sim 10^{-4}\, M_\oplus\,{\rm yr}^{-1}$, weakly dependent on time and orbital distance. The fraction of the pebble flux accreted by a single core is

$$f = \frac{\dot{M}_c}{\dot{M}_{\rm peb}} \approx 0.034 \left(\frac{\tau_f}{0.1}\right)^{-1/3} \left(\frac{M_c}{M_\oplus}\right)^{2/3} \left(\frac{r}{10\,{\rm AU}}\right)^{-1/2}, \tag{4.9}$$

which remains below unity until the core reaches its critical mass, ensuring that the pebble reservoir is not exhausted prematurely (Lambrechts & Johansen, 2014).

4.4 Physical Constraints and Mass Limits

4.4.1 Geometrical constraint: the 2D–3D transition

The vertical scale height of the gas disk and the Hill radius of the planetary core are given by Eqs. (2.11) and (3.12) respectively. Taking their ratio:

$$\frac{H}{r_H} = \frac{c_s/\Omega_K}{r\,(M_c/3M_*)^{1/3}} \tag{4.10}$$

$$= \frac{3^{1/3}}{\sqrt{G}} \sqrt{\frac{\gamma k_B}{\mu m_H}}\, M_*^{-1/6}\, M_c^{-1/3}\, r^{1/2}\, T^{1/2} \tag{4.11}$$

$$\equiv C\, r^{1/2}\, T^{1/2}, \tag{4.12}$$

with $C = (3^{1/3}/\sqrt{G})\sqrt{\gamma k_B/\mu m_H}\, M_*^{-1/6} M_c^{-1/3}$. The condition for three-dimensional pebble accretion is $r_H < H$, or equivalently $H/r_H > 1$. This imposes an upper limit on the core mass:

$$M_c < C'\,(rT)^{3/2}, \qquad C' = \frac{3}{G^{3/2}}\left(\frac{\gamma k_B}{\mu m_H}\right)^{3/2} M_*^{-1/2}. \tag{4.13}$$

When $r_H > H$, the pebble layer is thinner than the Hill sphere and accretion transitions from 3D to 2D, reducing the effective accretion rate. In PSʏCo, this geometrical limit is evaluated through the function `Mc_pebble_max` in `discGen_fd_pebble.py`, which implements Eq. (4.13) using the Hayashi temperature profile of Section 2.3.

4.4.2 Aerodynamical constraint: minimum pebble size in the Epstein regime

The friction time in the Epstein regime is $t_f = \rho_s a/\rho_g v_{\rm th}$ (Eq. 3.2). Using the midplane gas density $\rho_g = \Sigma_g \Omega_K/\sqrt{2\pi}c_s$ (Eq. 2.14) and the ratio $v_{\rm th}/c_s = \sqrt{8/\pi\gamma}$, the particle radius corresponding to a given Stokes number is

$$a = \frac{2}{\pi}\frac{\tau_f\,\Sigma_g}{\rho_s\sqrt{\gamma}}, \tag{4.14}$$

and the corresponding particle mass for a compact sphere ($m = \frac{4}{3}\pi\rho_s a^3$) is

$$m = \frac{32}{3\pi^2}\frac{\tau_f^3\,\Sigma_g^3}{\rho_s^2\,\gamma^{3/2}}. \tag{4.15}$$

These relations define the minimum particle mass required for solids to remain in the Epstein regime and participate in pebble accretion. In PSʏCo, Eq. (4.15) is implemented as the function `m_pebble_min`, with $\tau_f = 0.01$ and $\rho_s = 1.5\,{\rm g\,cm^{-3}}$ as default parameters. Pebble accretion is activated in the integrator when $m \geq m_{\rm threshold} = 10^{-3}$ g, corresponding to the onset of the `flag_peb` condition described in Chapter 5.

4.4.3 Pebble isolation mass: stopping condition

Pebble accretion ceases once the core reaches the *pebble isolation mass* $M_{\rm iso,peb}$. As derived in Section 2.3 from the condition $r_H \sim H$ with the Hayashi temperature profile (Eq. 2.21), and calibrated against hydrodynamical simulations (Lambrechts & Johansen, 2014):

$$M_{\rm iso,peb}(r) \approx 20\left(\frac{r}{5\,{\rm AU}}\right)^{3/4} M_\oplus. \tag{4.16}$$

This expression gives $\sim$10 $M_\oplus$ at 2 AU and $\sim$33 $M_\oplus$ at 10 AU. The pebble isolation mass plays a dual physical role: it marks the point at which pebble accretion is halted by the pressure gap, and simultaneously defines the critical core mass above which the gaseous envelope can no longer be sustained in hydrostatic equilibrium (Pollack et al., 1996; Lambrechts & Johansen, 2014). Once $M_c \geq M_{\mathrm{iso,peb}}$, the thermal support of the envelope is removed and rapid runaway gas accretion is triggered. In PSyCo, this condition is implemented through the function `Mc_pebble_iso` and activates the `flag_peb_iso` flag in the integrator, freezing the core mass and initiating the gas accretion phase described in Chapter 5.

5. Numerical Implementation in PSyCo

This chapter describes the numerical implementation of the pebble accretion module within PSyCo and presents the results of a validation run designed to characterize the behavior of the new mechanism. The implementation follows directly from the analytical prescription derived in Chapter 4 and is built upon the existing PSyCo integration framework, which handles disk evolution, solid and gas accretion, and planetary migration following Ida & Lin (2008). The central contribution of this work is the design and integration of a self-consistent pebble accretion module that operates alongside the classical Ida & Lin planetesimal accretion prescription, allowing a direct quantitative comparison between the two formation pathways within the same simulation environment. Additionally, Section 5.4 presents a systematic exploration of the effect of disk truncation on pebble accretion efficiency, examining four values of the tapering radius r_{taper} under fixed disk mass conditions.

5.1 The PSyCo Integration Framework

PSyCo evolves a set of planetary embryos forward in time within a protoplanetary disk, computing at each timestep the growth of solid cores, the onset and evolution of gas accretion, and the orbital migration driven by planet–disk interactions. The disk model follows the prescription of Section 2.4: the gas surface density decays exponentially as $\Sigma_g(r,t) \propto e^{-t/\tau_{\text{dep}}}$, with an exponential outer taper at r_{taper}. The solid surface density evolves proportionally through the dust-to-gas ratio, which depends on the stellar metallicity following Ida & Lin (2008).

The integration proceeds over a time array spanning from $t_{\text{min}} = 0.01$ yr to $t_{\text{max}} = \tau_{\text{dep}}$, with a fixed timestep of $\Delta t = 25$ yr. At each step, the code evaluates the local disk properties at the current orbital position of each embryo and updates its mass and orbital radius according to the active physical processes. The main loop structure can be summarized as four sequential blocks:

- **Block A** — Disk quantities and pebble accretion limits: evaluates $\Sigma_g(r,t)$, $m_{\text{peb,min}}$ (Eq. 4.15), and $M_{\text{iso,peb}}$ (Eq. 4.16) at the current position, and checks the activation condition for pebble accretion.

- **Block B** — Core mass update: computes the new core mass by integrating either the pebble accretion rate $\dot{M}_c$ (Eq. 4.6) or the classical Ida & Lin prescription, depending on the active flags.

- **Block C** — Stopping conditions: checks whether the core has reached $M_{\text{iso,peb}}$ or the classical isolation mass M_{iso}, and freezes the core mass accordingly.
- **Block D** — Gas accretion and migration: evaluates the onset of gas accretion via the Kelvin–Helmholtz timescale and updates the orbital radius under Type I or Type II migration.

5.2 The Pebble Accretion Module

The pebble accretion module introduced in this work operates through a set of physical flags that control the activation and deactivation of pebble accretion at each timestep and for each embryo independently. Table 5.1 summarizes the four flags and their physical meaning.

Table 5.1: *Physical flags governing the pebble accretion module in PSyCo. Each flag is evaluated independently for each planetary embryo at every timestep.*

Flag	Activation condition	Physical meaning
`flag_peb`	$m_{\text{peb,min}} \geq m_{\text{threshold}}$	Pebbles are large enough to be captured
`flag_peb_iso`	$M_c \geq M_{\text{iso,peb}}$	Core has opened a pressure gap; pebbles halted
`flag_hydro`	$M_c \geq M_{\text{hydro}}$	Envelope collapses; gas accretion begins
`flag_isog`	$M_{\text{planet}} \geq M_{g,\text{iso}}$	Gas isolation mass reached; accretion stops

5.2.1 Activation condition: `flag_peb`

At each timestep, the minimum pebble mass in the Epstein regime is evaluated using Eq. (4.15) with the local gas surface density $\Sigma_g(r,t)$, a fixed Stokes number $\tau_f = 0.01$, and an internal particle density $\rho_s = 1.5\,\mathrm{g\,cm^{-3}}$. Pebble accretion is activated when

$$m_{\text{peb,min}} \geq m_{\text{threshold}} = 10^{-3}\,\mathrm{g}, \tag{5.1}$$

which corresponds to the onset of the Epstein drag regime for particles of the relevant size. Since $m_{\text{peb,min}} \propto \Sigma_g^3$ and Σ_g decreases with orbital distance, this condition is easier to satisfy in the inner disk. In the validation run described in Section 5.3, `flag_peb` activates at $j = 1$ (the first timestep) for all four embryos. This is a direct consequence of the cubic scaling $m_{\text{peb,min}} \propto \Sigma_g^3$ (Eq. 4.15): at $t = 0$ the gas surface density has not yet been depleted by the exponential decay, and even at the outermost orbit (10 AU) the initial value $\Sigma_g \approx 26$ g cm^{-2} is sufficient to push $m_{\text{peb,min}}$ above the activation threshold $m_{\text{threshold}} = 10^{-3}$ g. The early activation therefore reflects the high initial gas content of the disk rather than any numerical artifact, confirming that pebble-sized particles are present throughout the disk at $t = 0$.

5.2.2 Core mass integration: Block B

When `flag_peb` is active and `flag_peb_iso` is not, the core mass is updated by integrating $\dot{M}_c$ (Eq. 4.6) over the timestep:

$$M_c^{(j)} = M_c^{(j-1)} + \dot{M}_c\left(M_c^{(j-1)},\, r,\, t,\, \Sigma_g^{(j)}\right)\, \Delta t, \tag{5.2}$$

with the constraint that $M_c^{(j)} \geq M_c^{(j-1)}$ (the core mass never decreases) and $M_c^{(j)} \leq M_{\mathrm{iso,peb}}$ (the core never exceeds the pebble isolation mass). In the current implementation, the pebble contribution is taken as the dominant growth mechanism when active, with the Ida & Lin prescription running in parallel as a reference. The total core mass at each step is taken as the maximum of the two:

$$M_c^{(j)} = \max\left(M_c^{\mathrm{pebble}},\, M_c^{\mathrm{IL}}\right), \tag{5.3}$$

where M_c^{IL} is the core mass predicted by the classical Ida & Lin prescription at the same timestep. This design decision ensures that the most efficient growth mechanism dominates at each orbital location and time. As discussed in Section 5.4, a physically more refined prescription, in which Ida & Lin growth proceeds first and pebbles contribute the excess, is identified as a priority for future work.

5.2.3 Stopping condition: `flag_peb_iso`

When the core mass reaches the pebble isolation mass,

$$M_c^{(j)} \geq M_{\mathrm{iso,peb}}(r) = 20\left(\frac{r}{5\,\mathrm{AU}}\right)^{3/4} M_\oplus, \tag{5.4}$$

`flag_peb_iso` is activated and the core mass is frozen at $M_{\mathrm{iso,peb}}$. Physically, this corresponds to the core having opened a pressure gap in the disk that halts the inward drift of pebbles (Section 4.4.3). From this point onward, the core no longer grows by solid accretion and the gas accretion phase begins if the hydrodynamic condition $M_c \geq M_{\mathrm{hydro}}$ is satisfied. The value of $M_{\mathrm{iso,peb}}$ is computed at the current orbital position and frozen once `flag_peb_iso` activates, so that subsequent migration does not alter the stopping mass.

5.3 Validation Run

To characterize the behavior of the pebble accretion module, we performed a deterministic validation run with four planetary embryos at fixed orbital positions of 2, 6, 8, and 10 AU, each with an initial mass of $M_0 = 0.01\, M_\oplus$, consistent with the lunar mass seed suggested by Lambrechts & Johansen (2012). The central star has $M_* = 1\, M_\odot$ and the disk mass is $M_d = 0.01\, M_\odot$, with a tapering radius of $r_{\text{taper}} = 70$ AU and a depletion timescale of $\tau_{\text{dep}} = 10$ Myr. These parameters are summarized in Table 5.2.

Table 5.2: *Initial conditions for the validation run.*

Parameter	Value	Description
M_*	$1\, M_\odot$	Stellar mass
M_d	$0.01\, M_\odot$	Disk mass
r_{taper}	70 AU	Outer tapering radius
τ_{dep}	10 Myr	Disk depletion timescale
Δt	25 yr	Integration timestep
M_0	$0.01\, M_\oplus$	Initial embryo mass
r_{obs}	2, 6, 8, 10 AU	Orbital positions
τ_f	0.01	Stokes number
ρ_s	$1.5\,\mathrm{g\,cm^{-3}}$	Particle density
$m_{\text{threshold}}$	10^{-3} g	Activation threshold

5.3.1 Pebble accretion timescales

Figure 5.1 shows the temporal evolution of the core mass $M_{c,\text{peb}}$ for each embryo, together with the accretion rate $\dot{M}_c$ (right axis) and the pebble isolation mass $M_{\text{iso,peb}}$ (dashed line). All four embryos reach $M_{\text{iso,peb}}$ within the disk lifetime, but on markedly different timescales that increase with orbital distance:

$$
\begin{aligned}
t_{\text{iso}}(2\,\text{AU}) &\approx 2.5 \times 10^4\,\text{yr},\\
t_{\text{iso}}(6\,\text{AU}) &\approx 1.5 \times 10^5\,\text{yr},\\
t_{\text{iso}}(8\,\text{AU}) &\approx 2.4 \times 10^5\,\text{yr},\\
t_{\text{iso}}(10\,\text{AU}) &\approx 3.7 \times 10^5\,\text{yr}.
\end{aligned} \tag{5.5}
$$

These timescales are two to three orders of magnitude shorter than the disk lifetime $\tau_{\text{dep}} = 10$ Myr, confirming that pebble accretion resolves the timescale problem discussed in Section 2.5. The growth curve at 2 AU is particularly striking: the core reaches $M_{\text{iso,peb}} = 10\, M_\oplus$ in only 10 active timesteps, reflecting the high gas surface density ($\Sigma_g \approx 328\,\mathrm{g\,cm^{-2}}$) and the strongly super-threshold pebble mass at that location. At wider orbits, the lower Σ_g produces slower but still efficient growth.

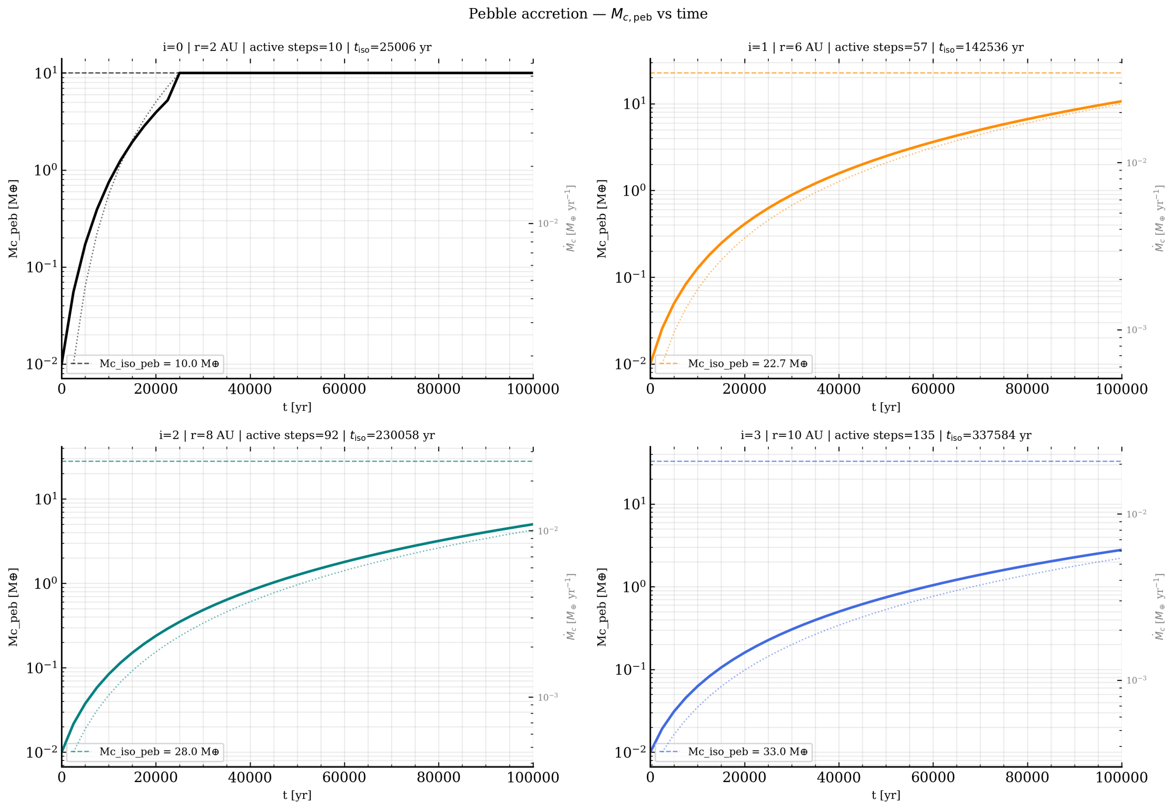

Figure 5.1: Temporal evolution of the core mass $M_{c,\text{peb}}$ (solid line, left axis) and the accretion rate $\dot{M}_c$ (dotted line, right axis) for four embryos at 2, 6, 8, and 10 AU. The dashed horizontal line marks the pebble isolation mass $M_{\text{iso,peb}}$ for each orbit. All embryos reach $M_{\text{iso,peb}}$ well within the disk lifetime. Panel titles indicate the number of active timesteps and the time at which $M_{\text{iso,peb}}$ is reached.

5.3.2 Comparison with classical accretion

Figure 5.2 compares the final core masses produced by pebble accretion with those obtained from the classical Ida & Lin prescription under identical initial conditions. At 2 AU, pebble accretion produces a core ∼27% more massive than the classical model (10.0 $M_\oplus$ vs 7.9 $M_\oplus$), reflecting the high efficiency of the Hill regime at small orbital separations where Σ_g is large.

At 6, 8, and 10 AU, however, the classical prescription produces significantly more massive cores (−82%, −81%, and −80% respectively relative to Ida & Lin). This result is directly linked to the role of $M_{\text{iso,peb}}$ as the stopping condition: once the core reaches the pebble isolation mass, solid accretion ceases entirely and the core remains frozen at that value. In the outer disk, $M_{\text{iso,peb}}$ ranges from 22.6 to 32.9 $M_\oplus$ (Table 5.3), while the Ida & Lin prescription continues growing the core to values exceeding 100 $M_\oplus$. The role of $M_{\text{iso,peb}}$ as a ceiling independent of disk structure is examined further in Section 5.4, where systematic variations in disk geometry allow us to disentangle the effect of local gas density from that of the stopping condition.

The validation run establishes the baseline behavior of the pebble accretion

Table 5.3: Pebble isolation mass $M_{\mathrm{iso,peb}}$ (Eq. 4.16) for the four orbital positions used in the validation run.

r [AU]	$M_{\mathrm{iso,peb}}$ [$M_\oplus$]
2	10.0
6	22.6
8	28.0
10	32.9

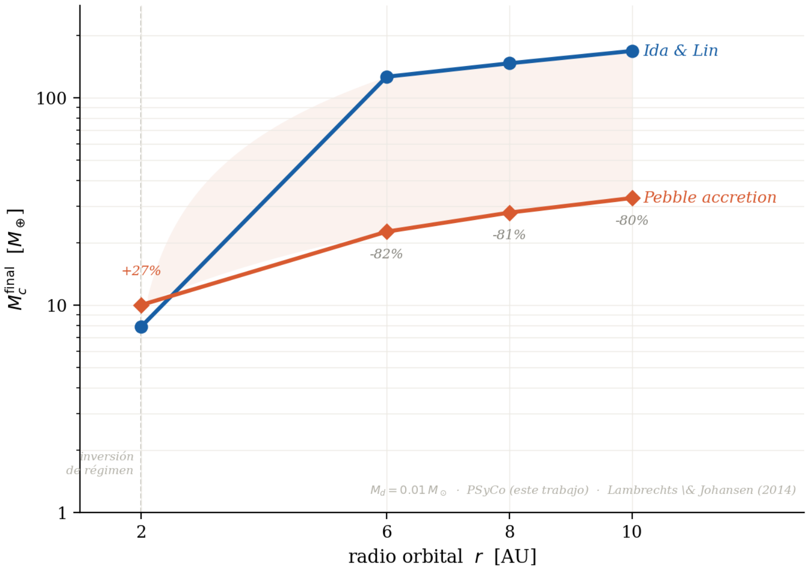

***Figure 5.2:** Final core mass M_c^{final} as a function of orbital radius for pebble accretion (coral) and the classical Ida & Lin prescription (blue), under identical initial conditions. Percentage labels indicate the relative difference $(M_c^{\mathrm{pebble}} - M_c^{\mathrm{IL}})/M_c^{\mathrm{IL}} \times 100\%$. At 2 AU, pebble accretion exceeds the classical result by 27% because the high local gas surface density ($\Sigma_g \approx 328$ g cm^{-2}) places the Hill regime fully in operation, yielding a large accretion cross-section. At 6, 8, and 10 AU the sign reverses: pebble accretion is less massive than the classical result because $M_{\mathrm{iso,peb}}$ acts as a hard ceiling that halts solid accretion once the core opens a pressure gap, while the Ida & Lin prescription continues growing the core indefinitely beyond that mass. The deficit therefore reflects a physical stopping condition, not an inefficiency of the mechanism.*

module under a standard disk configuration with $r_{\mathrm{taper}} = 70$ AU. The following section extends this analysis by varying the outer tapering radius of the disk, probing how the spatial concentration of gas mass affects the efficiency and timescale of core growth.

5.4 Disk Truncation and Core Growth Timescale

The validation run of Section 5.3 fixed the tapering radius at $r_{\mathrm{taper}} = 70$ AU. To examine how the radial extent of the gas disk affects pebble accretion, we repeated the integration for four values of the tapering radius, $r_{\text{taper}} \in \{5, 15, 30, 70\}$ AU, while keeping all other parameters identical to Table 5.2: the same disk mass $M_d = 0.01\,M_\odot$, the same four orbital positions (2, 6, 8, and 10 AU), and the same depletion timescale $\tau_{\text{dep}} = 10$ Myr. This design isolates the effect of disk geometry from that of disk mass: any difference between the four runs originates exclusively from where the gas surface density profile is cut off, not from how much gas the disk contains in total.

5.4.1 Normalization of the gas surface density

A subtlety in the disk model of Section 2.4 must be addressed before the truncated runs can be compared meaningfully. The gas surface density profile is normalized as

$$\Sigma_{g,0} = \frac{M_d}{2\pi\, r_0^q\, r_{\text{taper}}^{2-q}\, \Gamma(2-q)}, \tag{5.6}$$

where $r_0 = 1$ AU is a reference radius and q is the power-law index of the surface density profile. If r_{taper} is allowed to set this normalization directly, a smaller tapering radius produces a *larger* $\Sigma_{g,0}$, because the same disk mass M_d is redistributed over a smaller area. This would imply that a disk truncated at 5 AU is locally denser in its inner regions than a disk extending to 70 AU of the same total mass. Physically, a disk truncated by external photoevaporation does not redistribute its mass inward; it simply lacks gas beyond r_{taper}, while the inner disk retains the surface density set by the untruncated profile.

To avoid this artifact, the normalization $\Sigma_{g,0}$ is computed using a fixed reference value $r_{\text{taper,ref}} = 70$ AU for all four runs, while the exponential cutoff in $\Sigma_g(r, t)$ ((2.24)) still uses the tapering radius of the corresponding scenario:

$$\Sigma_g(r, t) = \Sigma_{g,0}\Big|_{r_{\text{taper,ref}}=70\,\text{AU}}\, f_g(t) \left(\frac{r}{1\,\text{AU}}\right)^{-q} \exp\left(-\frac{r}{r_{\text{taper}}}\right). \tag{5.7}$$

With this convention, all four disks share an identical surface density profile in the inner regions where $r \ll r_{\text{taper}}$, and differ only through the exponential cutoff that suppresses Σ_g as r approaches and exceeds r_{taper}. Figure 5.3 shows this behavior explicitly: the four curves coincide closely at small r and separate progressively as each profile approaches its own tapering radius, confirming that the corrected normalization isolates the geometric effect of truncation from any artificial change in disk mass concentration.

Table 5.4 lists Σ_g at $t = 0$ for the four orbital positions and four tapering radii under this corrected normalization.

For every orbital position, Σ_g increases monotonically with r_{taper}: the more extended disk supplies more gas locally at every radius, because the exponential cutoff suppresses the profile less. The effect strengthens with orbital distance, the ratio $\Sigma_g(70\,\text{AU})/\Sigma_g(5\,\text{AU})$ grows from 1.45 at 2 AU to 6.40 at 10 AU, since

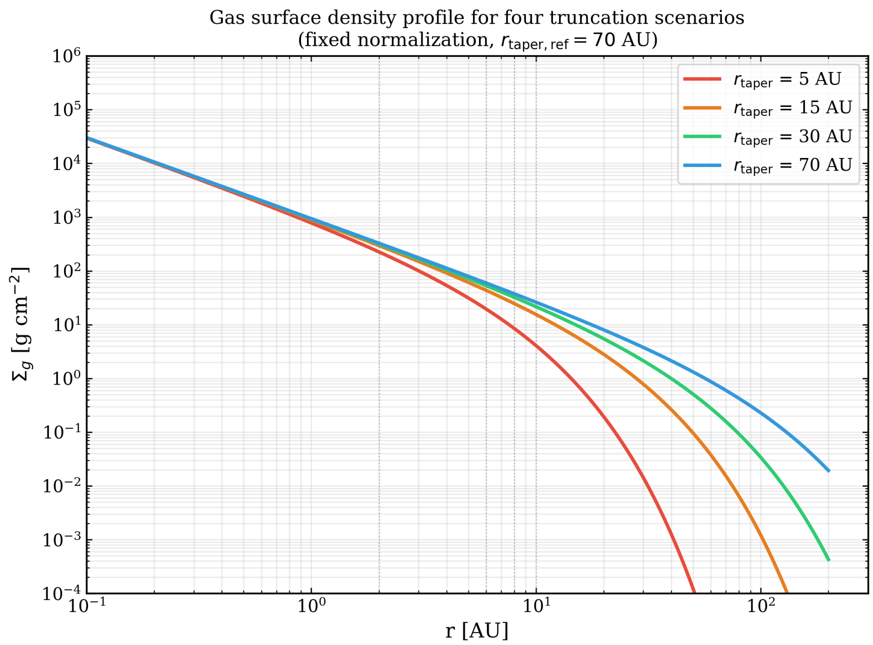

Figure 5.3: Gas surface density profile $\Sigma_g(r)$ at $t = 0$ for the four truncation scenarios (r_{taper} = 5, 15, 30, and 70 AU), computed with the fixed normalization of Eq. (5.7). All four curves coincide in the inner disk and diverge only near their respective tapering radii, where the exponential cutoff becomes dominant. Dotted vertical lines mark the orbital positions of the four embryos used in the validation run (2, 6, 8, and 10 AU).

Table 5.4: Gas surface density Σ_g [g cm^{-2}] at $t = 0$ for the four orbital positions and four tapering radii, computed with the fixed normalization of Eq. (5.7). All four columns share the same total disk mass $M_d = 0.01\ M_\odot$.

r_{taper} [AU]	Σ_g(2 AU)	Σ_g(6 AU)	Σ_g(8 AU)	Σ_g(10 AU)
5	226.0	19.5	8.5	4.1
15	295.1	43.5	24.7	15.5
30	315.4	53.1	32.3	21.6
70	327.6	59.5	37.6	26.1

outer orbits lie closer to the tapering radius itself and are therefore more sensitive to where the cutoff is placed.

5.4.2 Propagation to the accretion rate

Since the pebble accretion rate scales linearly with the local gas surface density, $\dot{M}_c \propto \Sigma_g$ (Eq. 4.6), the trend of Table 5.4 should propagate directly into $\dot{M}_c$. We verified this by comparing $\dot{M}_c$ across the four scenarios at the first timestep where pebble accretion is simultaneously active in all of them. At 2 AU, $\dot{M}_c(70\,\mathrm{AU})/\dot{M}_c(5\,\mathrm{AU}) \approx 1.9$, compared with $\Sigma_g(70\,\mathrm{AU})/\Sigma_g(5\,\mathrm{AU}) \approx 1.5$ at the same position; at 6 AU the ratios are ≈ 7.4 and ≈ 3.0, respectively. The accretion-rate ratio exceeds the surface-density ratio in both cases because $\dot{M}_c \propto M_c^{2/3}$ in addition to its linear dependence on Σ_g: the core in the more extended disk grows slightly faster from the start, which amplifies the gap through this feedback term as the integration proceeds. The ordering, however, is unambiguous and consistent at every orbital position probed: a more extended disk produces a higher local Σ_g and a correspondingly higher $\dot{M}_c$, at fixed total disk mass.

At 8 and 10 AU, the comparison cannot be extended to the $r_{\rm taper} = 5$ AU scenario using the same procedure, because $\dot{M}_c$ never becomes simultaneously active across all four runs at these orbital distances. This is not a numerical artifact but a direct consequence of the activation condition for pebble accretion (Eq. 5.1): since $m_{\rm peb,min} \propto \Sigma_g^3$, the steep suppression of Σ_g at $r_{\rm taper} = 5$ AU at these distances (Table 5.4) pushes $m_{\rm peb,min}$ below the activation threshold $m_{\rm threshold}$, and `flag_peb` never turns on. Physically, a disk truncated as compactly as 5 AU simply does not retain enough gas at 8 and 10 AU to sustain particles in the pebble-sized regime; the aerodynamic conditions required for efficient capture are not met. The truncation radius therefore does not only modulate the *rate* of pebble accretion , in the most compact scenario, it can suppress the mechanism entirely at sufficiently wide orbits.

Figure 5.4 shows the resulting core mass evolution $M_c(t)$ during the growth phase for the four embryos under the four truncation scenarios. The ordering of the curves at every orbital position mirrors the ordering of Σ_g in Table 5.4 and Figure 5.3: the most extended disk ($r_{\rm taper} = 70$ AU, blue) drives the fastest growth, while the most compact disk ($r_{\rm taper} = 5$ AU, red) is the slowest, consistent with the accretion-rate ratios quantified above. The separation between curves is negligible at 2 AU, where Σ_g itself is only mildly sensitive to $r_{\rm taper}$ (Table 5.4), and widens systematically toward 10 AU, where the $\Sigma_g(70\,\mathrm{AU})/\Sigma_g(5\,\mathrm{AU})$ contrast is largest. At 6, 8, and 10 AU, the missing red curve in the later panels reflects the absence of active pebble accretion for $r_{\rm taper} = 5$ AU discussed above, while the remaining three scenarios converge toward the same horizontal asymptote, set by $M_{\rm iso,peb}(r)$, on progressively longer timescales as $r_{\rm taper}$ decreases.

5.4.3 Implications for core growth

These results refine the interpretation of the deficit identified in Section 5.3.2 between pebble accretion and the classical Ida & Lin prescription at 6, 8, and 10 AU. That deficit was attributed to $M_{\rm iso,peb}$ acting as a hard ceiling, independent of the local gas supply. The truncation study presented here shows that this ceiling is reached *faster* in more extended disks, since a higher Σ_g drives a higher $\dot{M}_c$ at fixed orbital position. The pebble isolation mass itself, $M_{\rm iso,peb}(r) = 20\,(r/5\,\mathrm{AU})^{3/4}\,M_\oplus$ (Eq. 5.4), depends only on orbital distance through the disk aspect ratio and is therefore unaffected by $r_{\rm taper}$: all four scenarios converge to the same final core mass at a given orbital position, differing only in the time required to reach it. Disk truncation thus governs the timescale of pebble-driven core growth rather than its asymptotic outcome. A systematic quantification of $t_{\rm iso}(r_{\rm taper})$ across a broader parameter space is deferred to future work involving

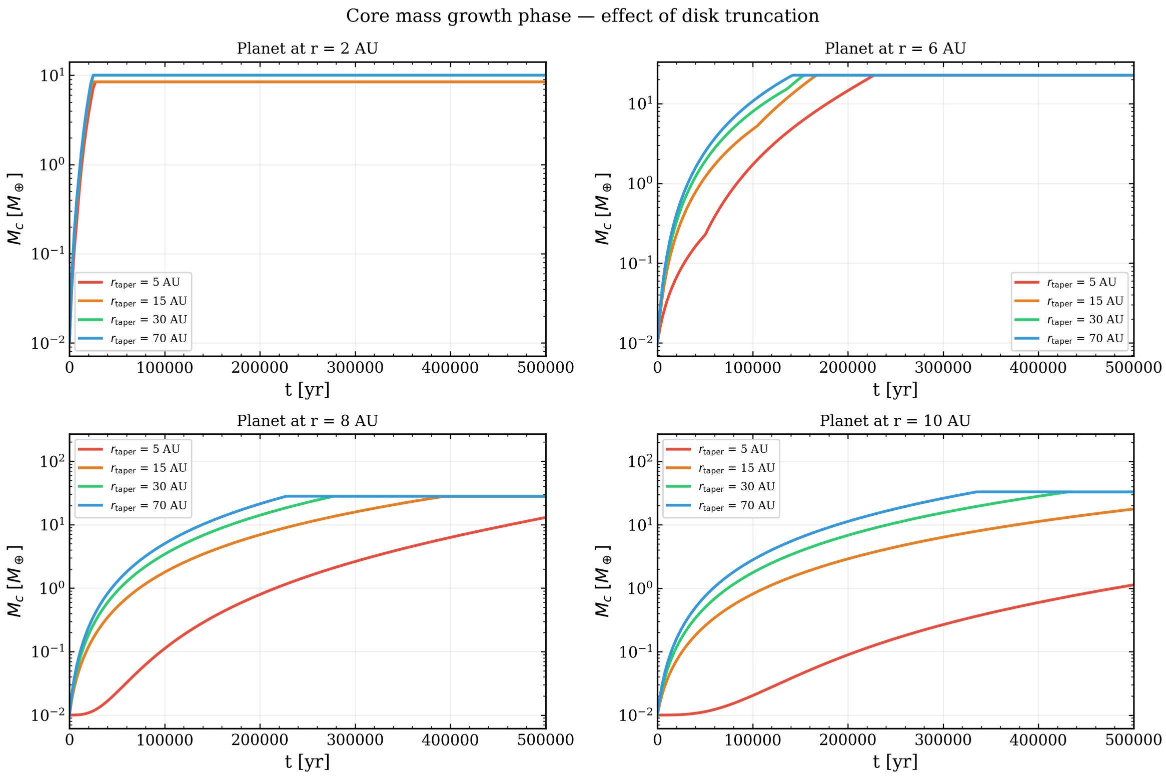

Figure 5.4: *Core mass evolution $M_c(t)$ during the pebble accretion growth phase for the four embryos (2, 6, 8, and 10 AU), under the four disk truncation scenarios of Table 5.4. The ordering of the curves, $r_{\rm taper}$ = 70 AU (blue) growing fastest, $r_{\rm taper}$ = 5 AU (red) growing slowest, follows directly from the Σ_g contrast shown in Figure 5.3. The separation between curves widens with orbital distance, and the $r_{\rm taper}$ = 5 AU curve is absent at 6, 8, and 10 AU where pebble accretion fails to activate.*

a full Monte Carlo population synthesis run, as outlined in Chapter 6.

6. Analysis & Future Work

6.1 Quantitative Analysis of Disk Truncation Effects on $t_{\rm iso}$

The truncation study of Section 5.4 established qualitatively that more extended disks drive faster core growth at fixed total disk mass. Here we quantify this behavior through the growth timescale $t_{\rm iso}(r, r_{\rm taper})$, defined as the time at which each embryo first reaches the pebble isolation mass $M_{\rm iso,peb}$ in the simulation. Table 6.1 summarizes the results for all four orbital positions and four truncation scenarios.

The values of $t_{\rm iso}$ across all cases where the isolation mass is reached range from $\sim 1.4 \times 10^5$ yr at 6 AU with $r_{\rm taper} = 70$ AU to $\sim 1.4 \times 10^6$ yr at 10 AU with $r_{\rm taper} = 5$ AU. These timescales are consistent with the analytical estimates of Ida et al. (2016), who find that the timescale for the pebble production front to reach the outer disk edge is generally of order 10^5 yr, well within the disk lifetime. The fact that our simulated $t_{\rm iso}$ values fall in this same range confirms that the PSyCo implementation of pebble accretion is operating in the physically expected regime.

At every orbital position where the isolation mass is reached, $t_{\rm iso}$ decreases monotonically with $r_{\rm taper}$ (Table 6.1, Figure 6.1). The physical origin of this trend follows directly from the normalization correction described in Section 5.4.1: since all four scenarios share the same total disk mass $M_d = 0.01\ M_\odot$ and the same inner normalization $\Sigma_{g,0}$, the exponential cutoff $\exp(-r/r_{\rm taper})$ in Eq. (5.7) is the sole source of difference between them. A more extended disk (large $r_{\rm taper}$) suppresses Σ_g less at the orbital positions of the embryos, producing a higher local gas surface density. Since $\dot{M}_c \propto \Sigma_g$ (Eq. 4.6), this translates directly into a higher accretion rate and a correspondingly shorter $t_{\rm iso}$. The effect strengthens with orbital distance: the ratio $t_{\rm iso}(r_{\rm taper} = 5\,{\rm AU})/t_{\rm iso}(r_{\rm taper} = 70\,{\rm AU})$ grows from ≈ 1.6 at 6 AU to ≈ 4.1 at 10 AU, reflecting the increasing sensitivity of Σ_g to the exponential cutoff at wider orbits.

Two qualitatively distinct failure modes appear in Table 6.1. At 2 AU with $r_{\rm taper} \leq 15$ AU, pebble accretion activates but growth remains incomplete: the disk disperses before the embryo closes the remaining gap to $M_{\rm iso,peb}$. At 8 and 10 AU with $r_{\rm taper} = 5$ AU, the mechanism is suppressed entirely because $m_{\rm peb,min}$ falls below $m_{\rm threshold}$ and `flag_peb` never activates (Section 5.4.2).

These results are broadly consistent with the findings of Qiao et al. (2023),

Table 6.1: Core growth timescale t_{iso} [yr] at which each embryo reaches the pebble isolation mass $M_{\text{iso,peb}}$, for the four orbital positions and four disk truncation scenarios. Dashes indicate cases where $M_{\text{iso,peb}}$ was not reached within the disk lifetime τ_{dep} = 10 Myr; the two distinct failure modes are discussed in the text.

r [AU]	$r_{\text{taper}} = 5$ AU	$r_{\text{taper}} = 15$ AU	$r_{\text{taper}} = 30$ AU	$r_{\text{taper}} = 70$ AU
2	—	—	2.50×10^4	2.50×10^4
6	2.27×10^5	1.67×10^5	1.55×10^5	1.42×10^5
8	6.12×10^5	3.92×10^5	2.77×10^5	2.27×10^5
10	1.39×10^6	6.67×10^5	4.30×10^5	3.35×10^5

who demonstrated that external photoevaporation of the outer disk limits the pebble mass reservoir available for core growth, and that sufficiently strong irradiation can suppress planetary growth at wide orbital separations. The key methodological difference is that Qiao et al. (2023) model the disk truncation as a dynamical process driven by a time-varying FUV radiation field, whereas this work treats r_{taper} as a fixed geometric parameter. This approach isolates the effect of disk extent from the time-dependence of the photoevaporative process, allowing a clean quantification of how the final spatial scale of the disk governs the efficiency of pebble-driven core growth. The two approaches are complementary: our results establish the $t_{\text{iso}}(r_{\text{taper}})$ dependence in a controlled setting, providing a baseline against which more dynamically realistic truncation models can be compared.

6.2 Interaction Between Pebble Accretion and the Ida & Lin Prescription

In the current implementation, the total core mass at each timestep is taken as the maximum of the pebble accretion contribution and the classical Ida & Lin prescription (Eq. 5.3). This design decision ensures that the most efficient growth mechanism dominates at each orbital location and time, but it carries an important limitation: it does not model a physical competition between the two mechanisms, but rather selects one over the other at each step without allowing them to operate simultaneously or sequentially in a physically motivated way.

The consequence of this approach becomes apparent at 2 AU, where the Ida & Lin prescription reaches $M_{\text{iso,peb}}$ before pebble accretion alone can do so. In that case, the `max()` operation attributes the growth to pebble accretion in the combined array `Mc_peb`, even though the physical agent completing the growth is the planetesimal prescription.

A physically more motivated approach would allow the two mechanisms to contribute additively rather than competitively. Under such a prescription, the Ida & Lin planetesimal accretion would build the core to its classical isolation mass first, and pebble accretion would then contribute the excess mass up to $M_{\text{iso,peb}}$. This sequential model would better reflect the temporal ordering of the

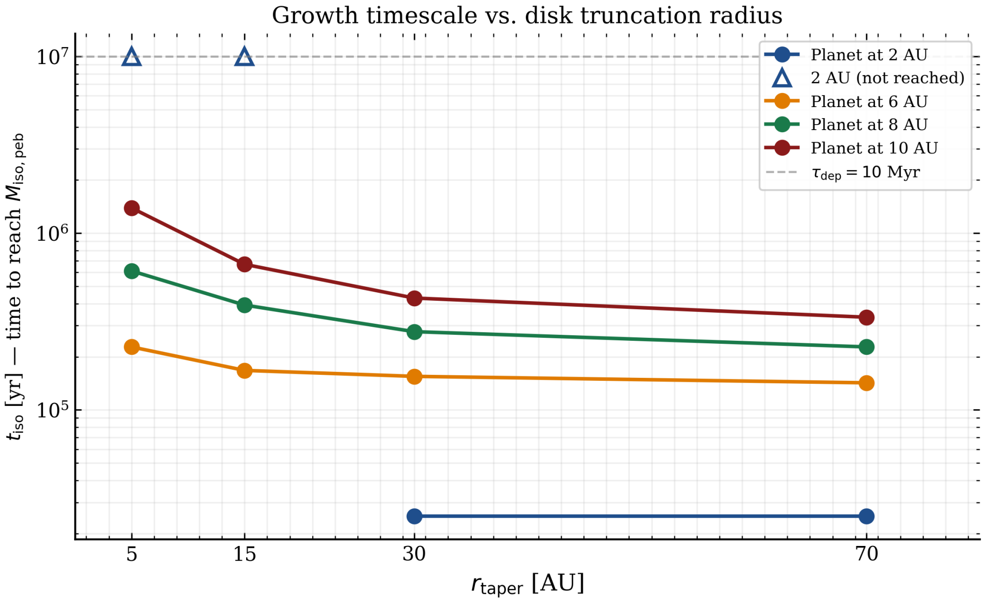

Figure 6.1: Core growth timescale t_{iso} as a function of the disk tapering radius r_{taper} for four planetary embryos at 2, 6, 8, and 10 AU. Filled circles connected by solid lines mark cases where the embryo reached $M_{iso,peb}$ within τ_{dep} = 10 Myr; open triangles at 10^7 yr indicate cases where the isolation mass was not reached. At every orbital position, t_{iso} decreases monotonically with r_{taper}. The embryo at 2 AU fails to complete growth for $r_{taper} \leq 15$ AU due to disk dispersal, while the absence of points at 8 and 10 AU for r_{taper} = 5 AU reflects complete suppression of pebble accretion. The dashed horizontal line marks τ_{dep}.

two processes, planetesimal accretion dominates in the early, gas-rich phase, while pebble accretion becomes increasingly important as the pebble flux builds up, and would produce a more gradual and physically transparent transition between regimes. Implementing this additive prescription is identified as a priority for future development of the PSyCo pebble accretion module.

6.3 Future Work

The results presented in this thesis establish a working implementation of pebble accretion within PSyCo and demonstrate its physical behavior across a range of disk truncation scenarios. Several directions for future development follow naturally from the limitations identified above.

The most immediate priority is the recalibration of the gas accretion phase. The Kelvin–Helmholtz timescale currently implemented in PSyCo produces gas masses that significantly exceed physical expectations for giant planets, reaching values of order 10^3–10^4 $M_\oplus$ in some cases (Ida & Lin, 2008). A more physically realistic prescription would incorporate the pebble isolation mass $M_{iso,peb}$ as the critical core mass that triggers envelope collapse, replacing the current hydrodynamic threshold M_{hydro}. This modification would ensure that the onset of rapid gas accretion is consistently tied to the pebble accretion stopping condition, producing gas giant masses in agreement with the observed population (Pollack et al., 1996).

A second priority is the extension of the truncation study to a broader parameter space. The four scenarios examined in Chapter 5 fix the disk mass at $M_d = 0.01\ M_\odot$ and vary only $r_{\rm taper}$. A natural extension would explore the joint dependence of $t_{\rm iso}$ on $r_{\rm taper}$, M_d, and the stellar metallicity Z_0, which controls the dust-to-gas ratio and therefore the solid surface density available for planetesimal accretion. This two-dimensional parameter space could be sampled efficiently with a Monte Carlo population synthesis run, in which $r_{\rm taper}$ and M_d are drawn from observed distributions of disk sizes and masses in young stellar clusters (Qiao et al., 2023).

Finally, the PSyCo framework is currently limited to a fixed stellar mass of $M_* = 1\ M_\odot$. Extending the pebble accretion module to sub-solar and super-solar mass stars would allow a direct comparison with the observed dependence of planetary system architectures on stellar mass (Lambrechts & Johansen, 2014), and would make PSyCo suitable for statistical studies of planetary populations across a range of host star properties. These extensions, taken together, would transform the current implementation from a validation tool into a fully operational instrument for planetary population synthesis in truncated disk environments.

7. Conclusions

The timescale problem in planet formation, the discrepancy between classical core growth timescales and the observed lifetime of protoplanetary disks, provided the central motivation for this work. The problem is sharpest in the outer disk: at orbital distances beyond $\sim$5 AU, planetesimal accretion predicts formation timescales of 10^7–10^8 yr, yet gas giants and directly imaged planets are observed at these separations, most notably HR 8799 at 14–68 AU (Marois et al., 2010) and β Pic b at $\sim$10 AU (Lagrange et al., 2010). Something must grow cores faster. This work addressed that problem by designing, implementing, and validating a pebble accretion module within PSyCo, the planetary population synthesis code developed at Universidad de Antioquia, based on the analytical prescription of Lambrechts & Johansen (2014). The module operates alongside the existing Ida & Lin planetesimal accretion prescription, enabling a direct quantitative comparison between the two formation pathways under identical disk conditions within the same simulation environment.

The validation run with four planetary embryos at fixed orbital positions of 2, 6, 8, and 10 AU confirmed that pebble-driven cores reach the pebble isolation mass $M_{\rm iso,peb}$ on timescales ranging from 2.5×10^4 yr at 2 AU to 3.7×10^5 yr at 10 AU. These values are two to three orders of magnitude shorter than the classical Ida & Lin prescription under identical conditions, and are consistent with the analytical timescale estimates of Lambrechts & Johansen (2012). This result confirms that the PSyCo implementation of pebble accretion operates in the physically expected regime and resolves the timescale problem for all four orbital positions probed. This confirmation is not merely a numerical check: it establishes that PSyCo can now serve as a platform for population synthesis studies in which pebble accretion is the dominant core growth mechanism. The efficiency of the mechanism at wide orbital separations opens the possibility of forming not only gas giant cores but also super-Earth mass bodies well within the disk lifetime.

The central scientific contribution of this work is the systematic study of disk truncation as a control parameter for pebble-driven core growth. By varying the outer tapering radius $r_{\rm taper} \in \{5, 15, 30, 70\}$ AU at fixed total disk mass $M_d = 0.01\ M_\odot$, this study isolates the geometric effect of disk extent from variations in disk mass or pebble flux. The results demonstrate that $r_{\rm taper}$ governs the *timescale* of pebble-driven core growth rather than its asymptotic outcome: all scenarios that complete growth converge to the same final core mass set by $M_{\rm iso,peb}(r)$, which depends only on orbital distance and is independent of disk geometry. More extended disks supply greater local gas surface density Σ_g at every orbital position, driving higher accretion rates $\dot{M}_c \propto \Sigma_g$ and correspondingly shorter $t_{\rm iso}$. The quantitative dependence, summarized in Table 6.1 and Figure 6.1, shows that the ratio $t_{\rm iso}(r_{\rm taper} = 5\,\mathrm{AU})/t_{\rm iso}(r_{\rm taper} = 70\,\mathrm{AU})$ grows from ≈ 1.6 at 6 AU to ≈ 4.1 at 10 AU, reflecting the increasing sensitivity of Σ_g to the exponential cutoff at wider orbits.

In the most compact configuration ($r_{\rm taper} = 5$ AU), two physically distinct failure modes emerge. At 8 and 10 AU pebble accretion is completely suppressed, while at 2 AU the mechanism activates but growth remains incomplete before disk dispersal. As discussed in Section 6.1, this distinction matters for interpreting observed planetary populations in compact disk environments: the absence of a giant planet at wide orbital separation may indicate not that pebble accretion failed, but that the disk was truncated before the isolation mass could be reached.

Several aspects of the implementation require further development before PSyCo can be used for full statistical population synthesis studies. The most pressing are the recalibration of the Kelvin–Helmholtz gas accretion timescale, which currently produces unrealistically massive envelopes, and the replacement of the `max(pebble, IL)` core mass combination with a physically motivated additive prescription in which planetesimal accretion builds the core to its classical isolation mass first and pebble accretion contributes the excess up to $M_{\rm iso,peb}$. The extension of the truncation study to a broader parameter space, varying M_d, Z_0, and M_* in addition to $r_{\rm taper}$, and the coupling with a dynamical photoevaporation model following Qiao et al. (2023) and Herrera et al. (2026), represent the natural next steps toward a complete population synthesis tool for planet formation in truncated disk environments. With these developments, PSyCo would become a fully operational instrument for studying how disk geometry, shaped by the stellar environment at birth, ultimately governs the architecture of planetary systems.

A. Appendix: PSyCo

Overview and Authorship

PSyCo (Planetary Synthesis Code) is a planetary population synthesis code developed by the Solar, Earth, and Planetary Physics (SEAP) group at FACom, Instituto de Física, Universidad de Antioquia. The code evolves planetary embryos forward in time within a protoplanetary disk, computing solid-core growth, gas accretion, and orbital migration self-consistently under the prescriptions of Ida & Lin (2008). The original implementation was developed collaboratively by Sebastián Hernández, Germán Chaparro, Pablo Cuartas, Sofía Arboleda, and Santiago Herrera. The source code is publicly available at `https://github.com/Bang1329/PsyCo/tree/main/Main`.

The contribution of the present work consists of the design and integration of a self-consistent pebble accretion module within the existing PSyCo framework, implemented in the files `discGen_fd_pebble.py` and `PsyCo_fd_pebble.py`. All other components of PSyCo — disk evolution, gas accretion, and migration — are prior work by the SEAP group and are used here without modification.

New Functions in `discGen_fd_pebble.py`

The pebble accretion module introduced in this work adds four functions to the disk generation module:

- `Mcore_pebble_dot`(M_c, r, t, Σ_g, M_*, Z_0): returns the instantaneous core mass accretion rate $\dot{M}_c$ in the Hill regime (Eq. 4.6), following Lambrechts & Johansen (2014). Valid for $\tau_f \sim 0.01$–0.1 and $M_c > M_t$.

- `Mc_pebble_iso`(r, M_*, L_*): returns the pebble isolation mass $M_{\text{iso,peb}}(r)$ (Eq. 4.16).

- `Mc_pebble_max`(r, M_*, L_*): returns the geometrical upper limit on core mass for 3D pebble accretion (Eq. 4.13).

- `m_pebble_min`(Σ_g, τ_f, ρ_s): returns the minimum pebble mass in the Epstein drag regime (Eq. 4.15), used to evaluate `flag_peb` at each timestep.

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